The Pol-InSAR Course

Synthetic Aperture Radar (SAR)

Part 4: Differential InSAR

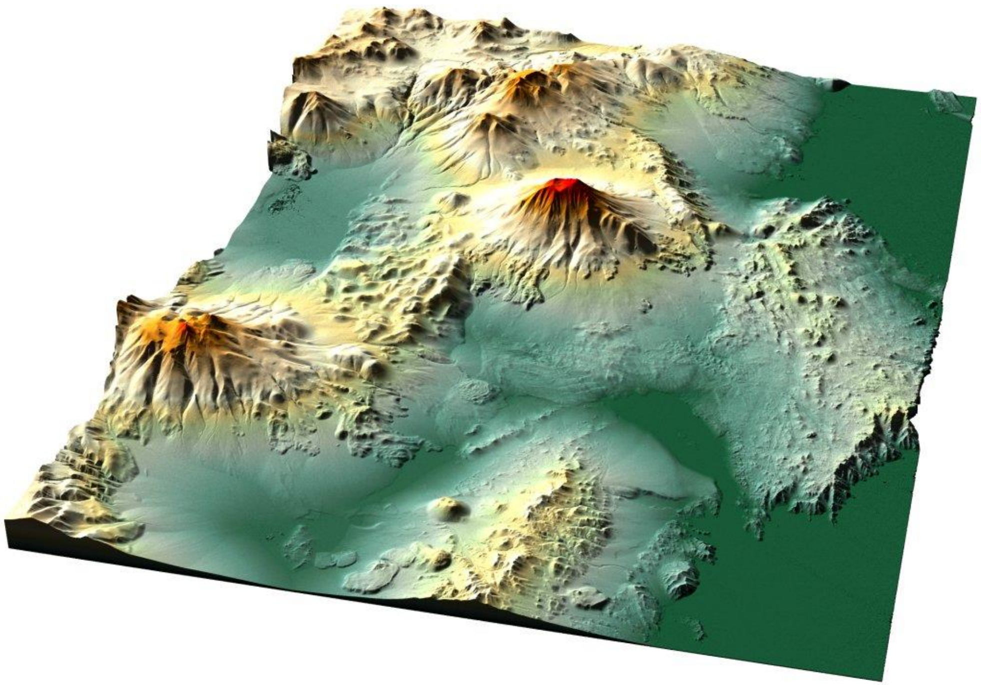
Prepared by DLR-HR's Pol-InSAR Team

German Aerospace Center (DLR), Microwaves & Radar Institute (HR), Pol-InSAR Research Group

Email: pau.prats@dlr.de, kristina.belinska@dlr.de, kostas.papathanassiou@dlr.de

DLR esa ESAMAAP EEBIOMASS

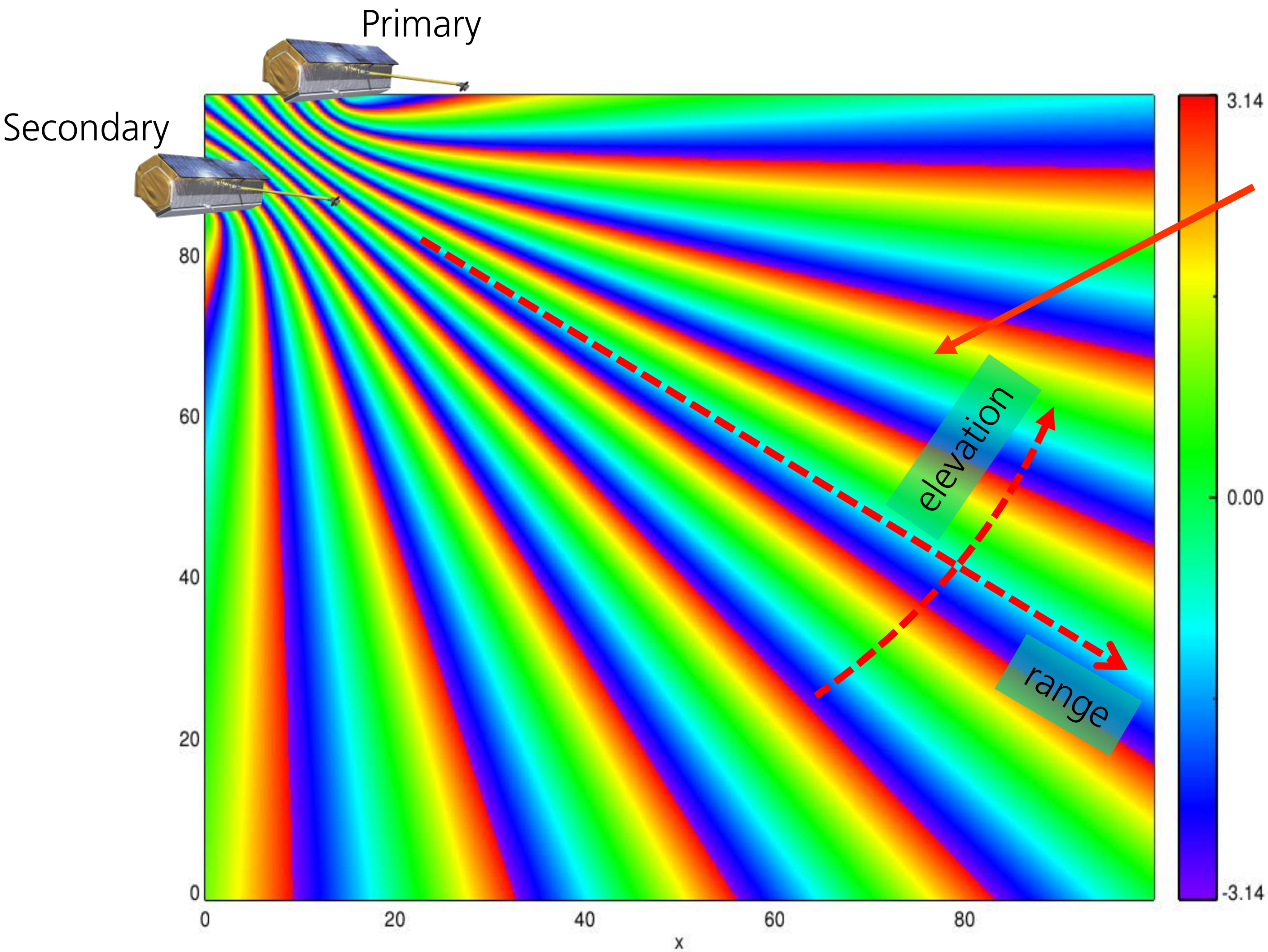
What can we do with SAR Interferometry?



Digital Elevation Model (DEM) over the Atacama desert, Chile, computed with two single-pass acquisitions acquired by TanDEM-X.

Massonnet et al., “The displacement field of the Landers earthquake mapped by radar interferometry,” Nature 364, 1993.

Cross-Track SAR Interferometry (revisited)



$$\varphi = \frac{4\pi}{\lambda} \cdot (r_2 - r_1) = \frac{4\pi}{\lambda} \Delta r$$

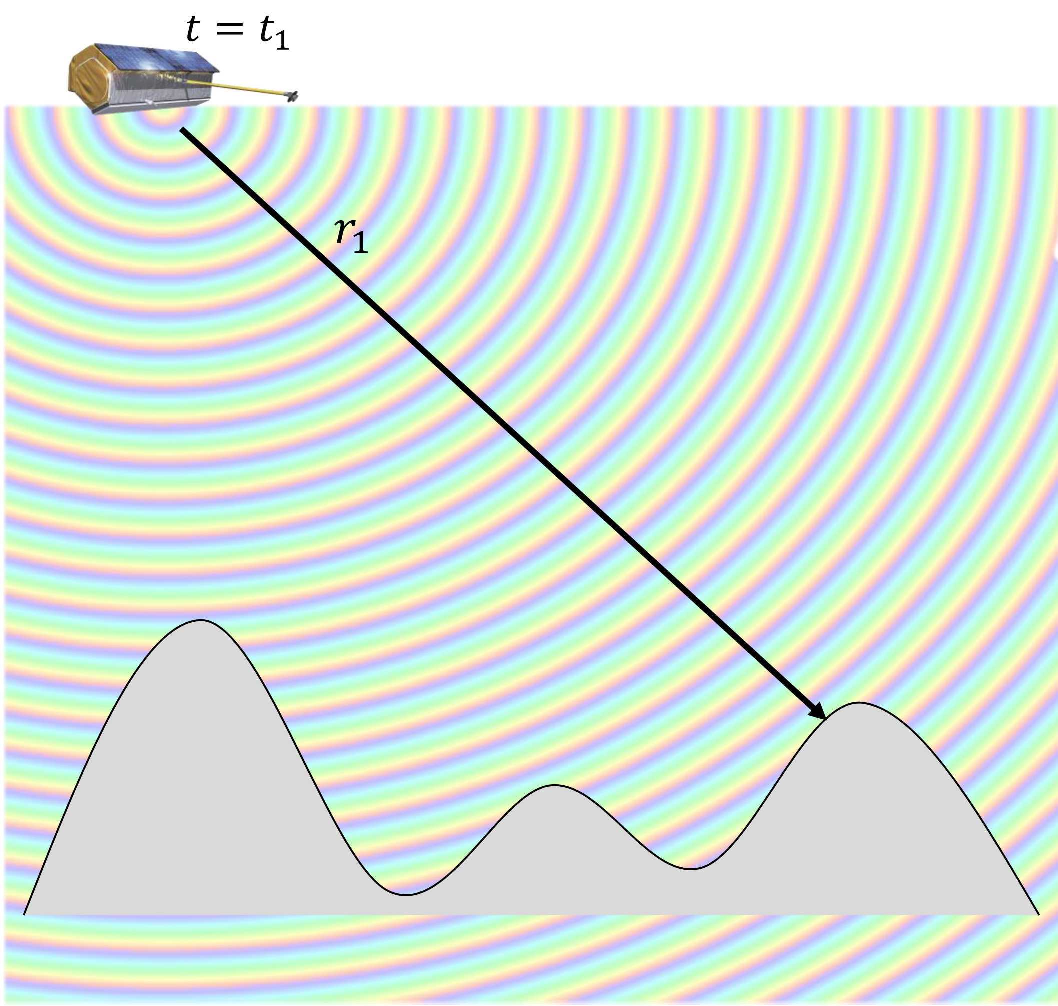
with Δr being the different distance to the target from each sensor position

$$\varphi_{topo} = k_z \cdot h$$

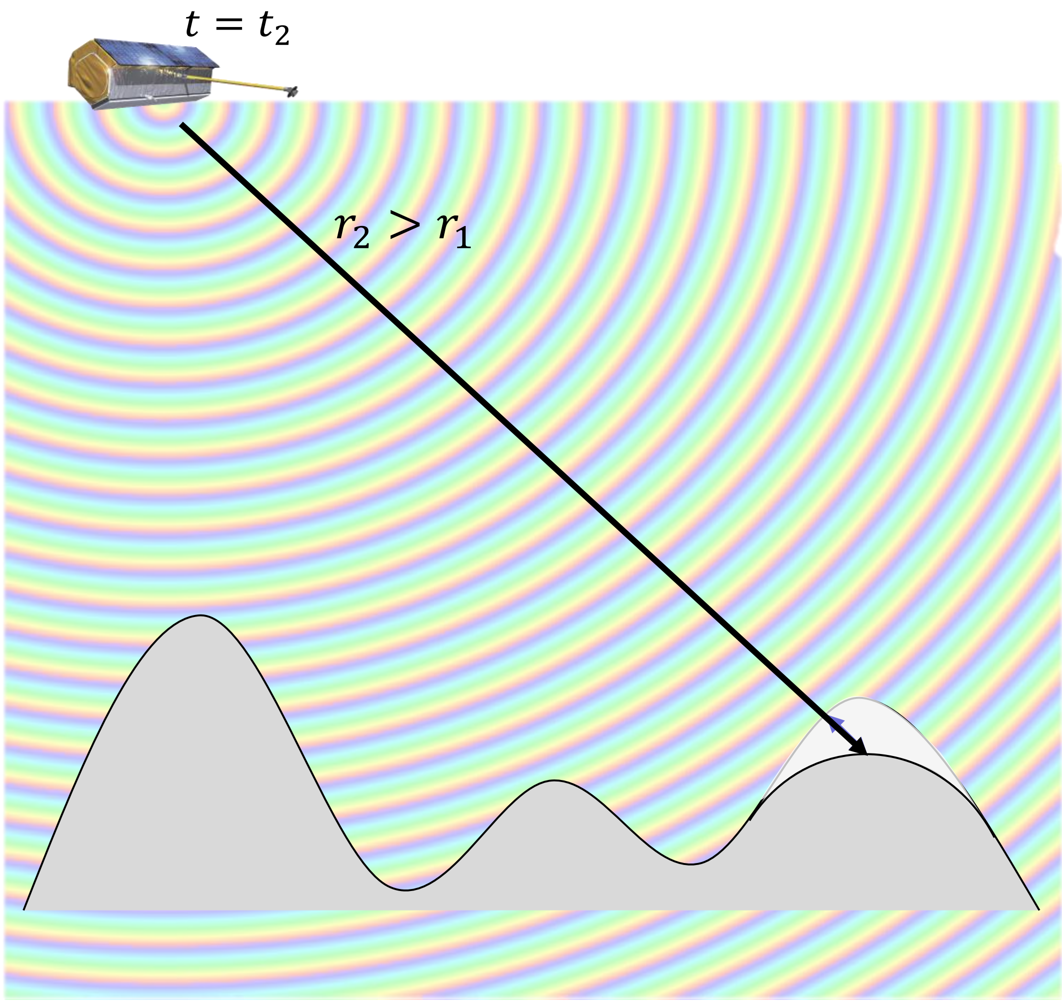
with

$$k_z = -\frac{4\pi}{\lambda} \frac{B \cdot \cos(\theta - \alpha)}{r \cdot \sin\theta}$$

Differential SAR Interferometry Principle



Differential SAR Interferometry Principle

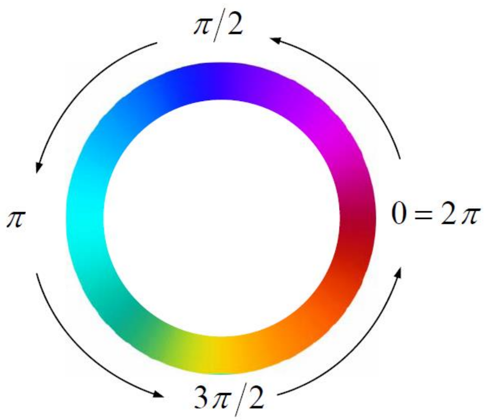


$$\varphi = \frac{4\pi}{\lambda} \cdot (r_2 - r_1) = \frac{4\pi}{\lambda} \Delta \mathrm{r}$$

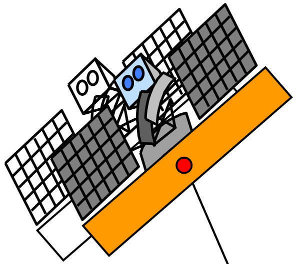
with Δr being the motion of the target in the **looking direction** of the radar (line of sight)

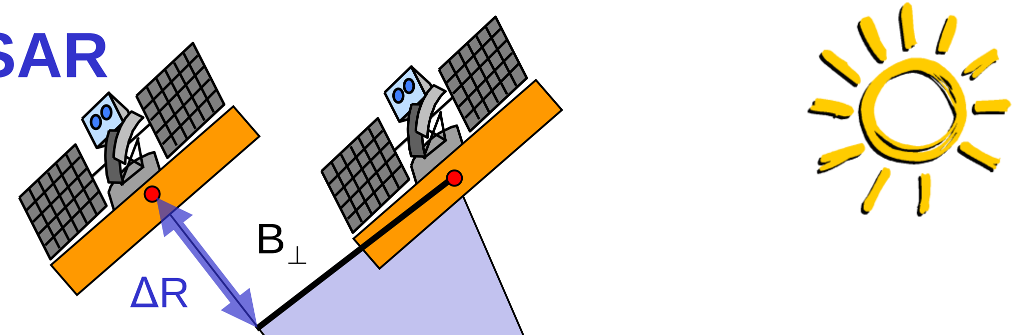
$$\varphi = 2\pi \rightarrow \Delta r_{2\pi} = \frac{\lambda}{2}$$

$$1 \text{ fringe} \equiv \frac{\lambda}{2}!!!$$



D-InSAR vs InSAR





<u>Example ERS</u>: Space-borne C-band (λ=0.056m) interferometer with incidence θ=23° at range R=870Km

1 phase cycle of 2π (i.e., 1 fringe) corresponds to:

D-InSAR

$$\sigma_R = \frac{\lambda}{4\pi}\sigma_\varphi = \frac{\lambda}{4\pi}2\pi = 0.028 \text{ m}$$

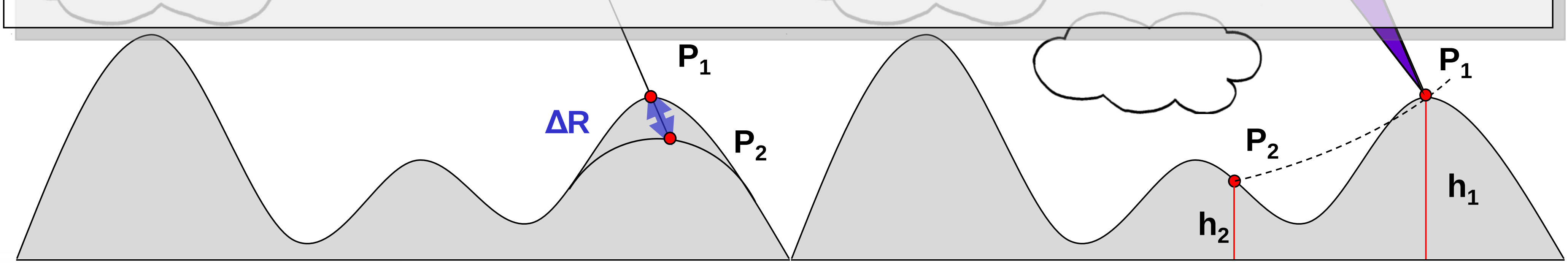
(in LOS)

InSAR

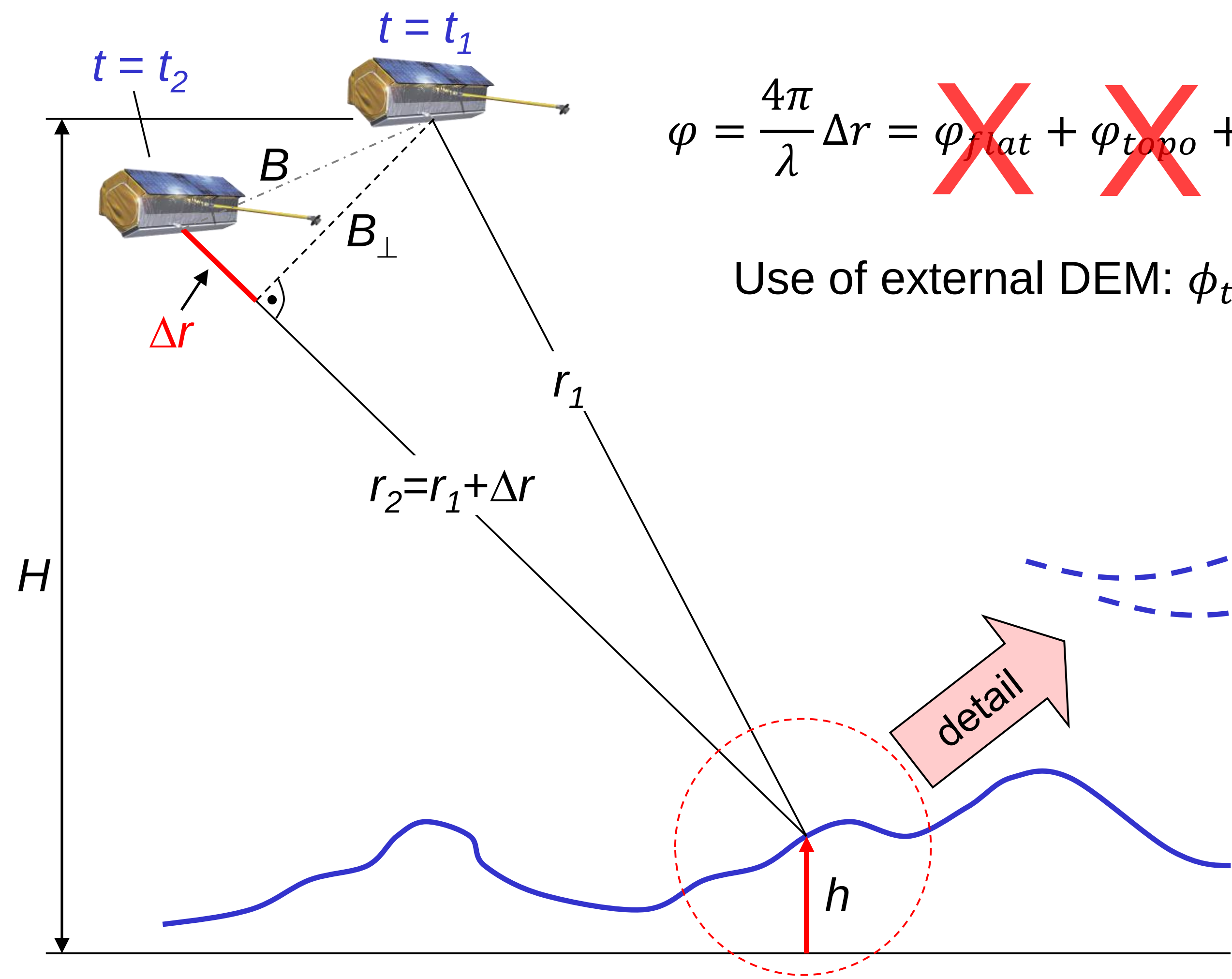
$$\sigma_z = \frac{\lambda}{4\pi}\frac{R\sin(\theta)}{B_\perp}\sigma_\varphi = \frac{\lambda}{4\pi}\frac{R\sin(\theta)}{B_\perp}2\pi$$

For a $B_\perp$=100m baseline : σ_z=100m terrain elevation

For a $B_\perp$=200m baseline: σ_z= 50m terrain elevation

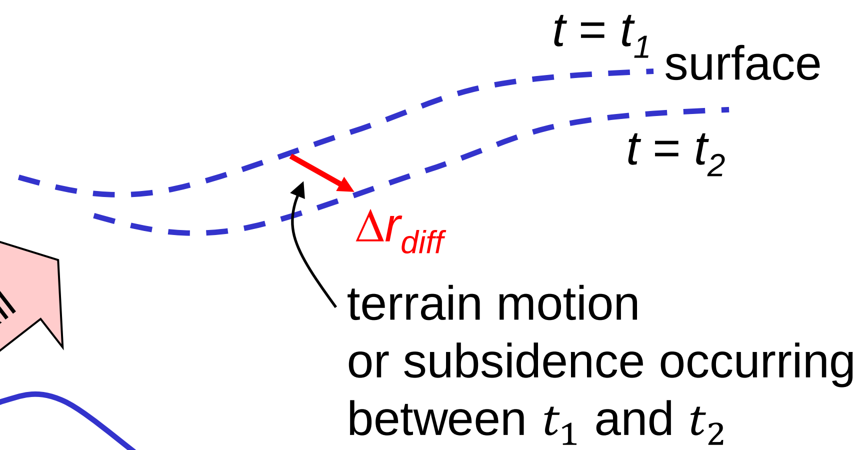


Differential SAR Interferometry Principle



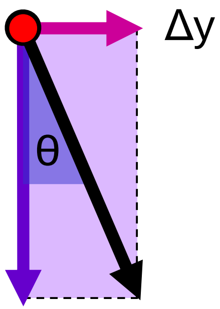
$$\varphi = \frac{4\pi}{\lambda}\Delta r = \cancel{\varphi_{flat}} + \cancel{\varphi_{topo}} + \varphi_{diff} \;\rightarrow\; \varphi_{diff} = \frac{4\pi}{\lambda}\Delta r_{diff}$$

Use of external DEM: $\phi_{topo} = k_z \cdot (h - h_{mean})$

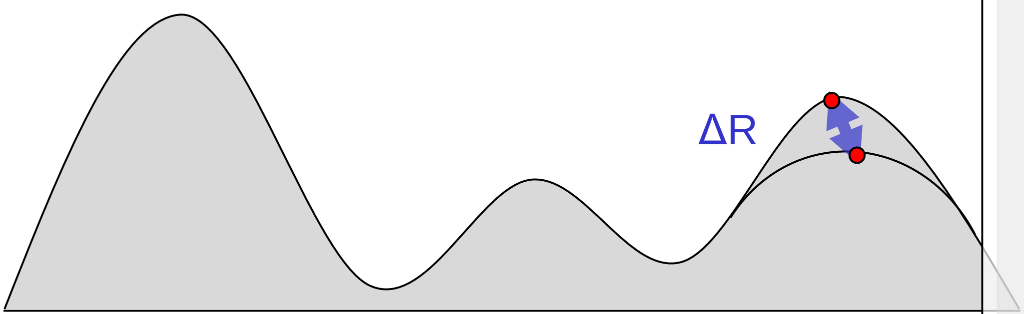


Projection to Vertical or Horizontal

Ground range (horizontal displacement) &
Height (vertical displacement):



$$\Delta R = \Delta y \sin(\theta) + \Delta z \cos(\theta)$$



Example ERS: Space-borne C-band (λ=0.056m) interferometer with incidence θ=23° at range R=870Km. 1 phase cycle of 2π (i.e. 1 fringe) corresponds to:

$$\sigma_z = \frac{1}{\cos(\theta)}\sigma_R = 0.030\ \mathrm{m} \qquad \text{(vertical)}$$

$$\sigma_y = \frac{1}{\sin(\theta)}\sigma_R = 0.072\ \mathrm{m} \qquad \text{(horizontal)}$$

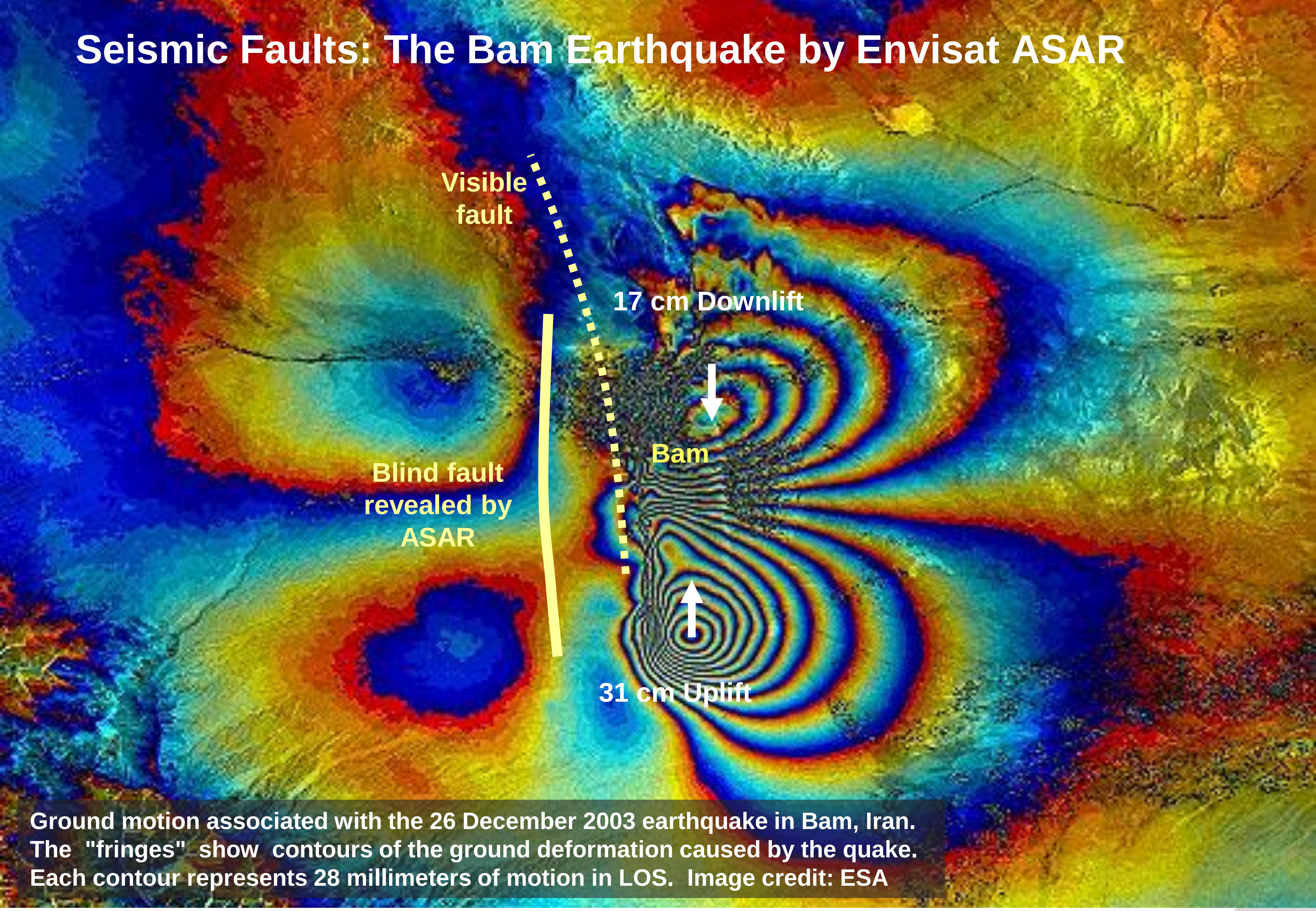
Assuming the ability to measure the interferometric phase with an accuracy of 20°:

$$\sigma_R = \frac{\lambda}{4\pi}\sigma_\varphi = \frac{\lambda}{4\pi}\frac{20}{360}2\pi = 0.0015\ \mathrm{m} \quad \text{(in LOS)}$$

$$\sigma_z = \frac{1}{\cos(\theta)}\sigma_R = 1.6\ \mathrm{mm} \qquad \text{(vertical)}$$

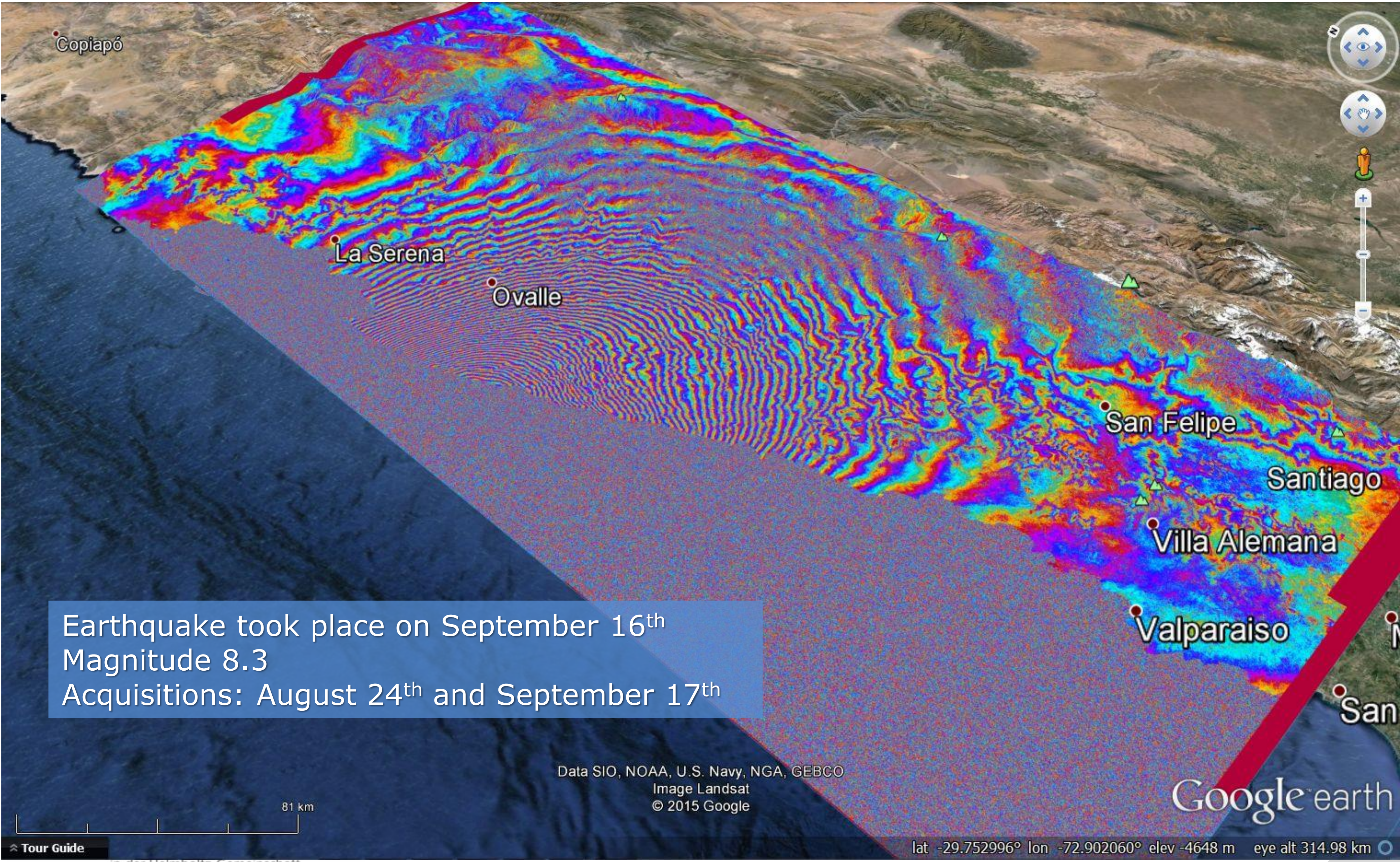
$$\sigma_y = \frac{1}{\sin(\theta)}\sigma_R = 4.0\ \mathrm{mm} \quad \text{(horizontal)}$$

Seismic Faults: The Bam Earthquake by Envisat ASAR

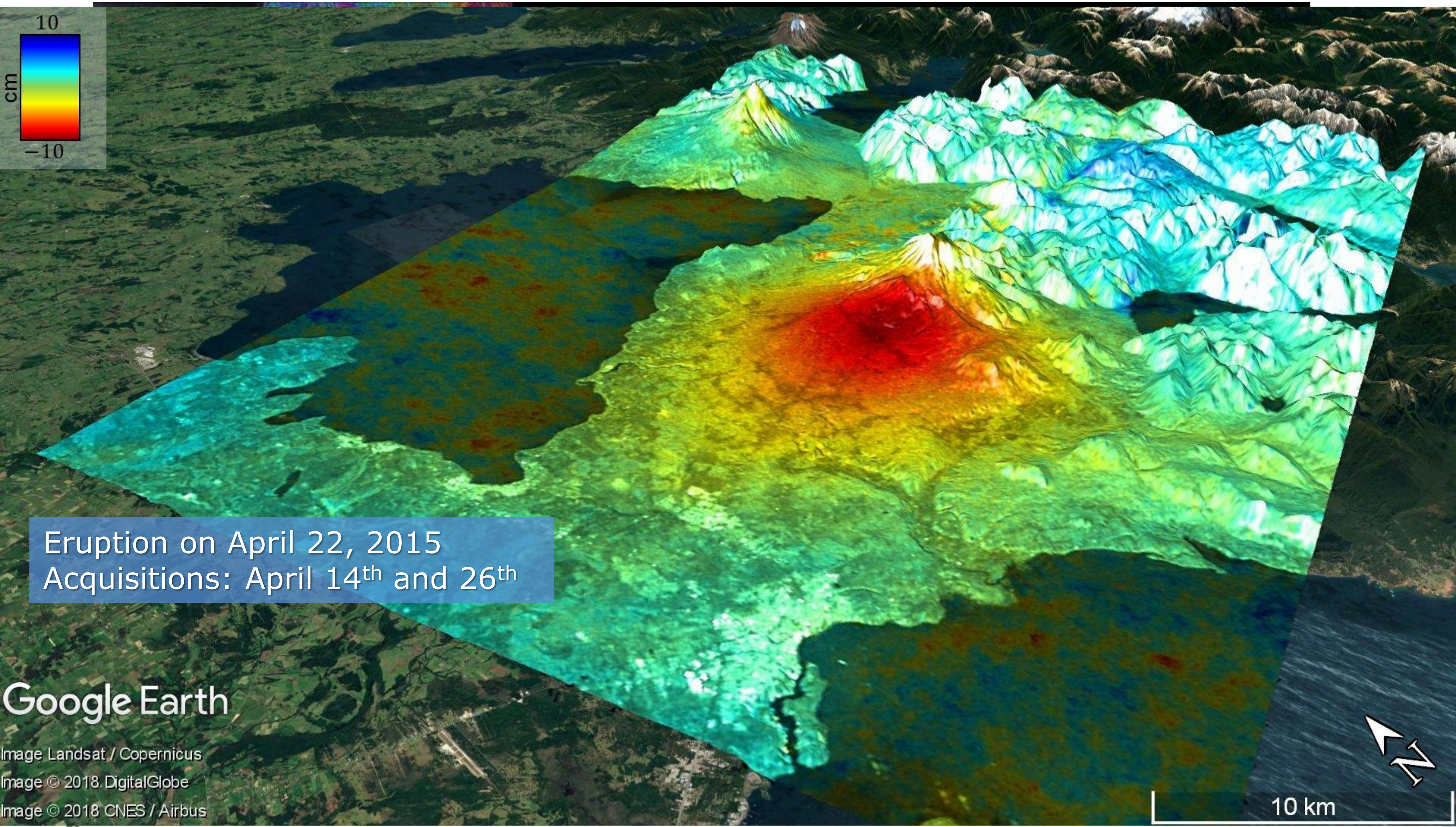


Ground motion associated with the 26 December 2003 earthquake in Bam, Iran. The "fringes" show contours of the ground deformation caused by the quake. Each contour represents 28 millimeters of motion in LOS. Image credit: ESA

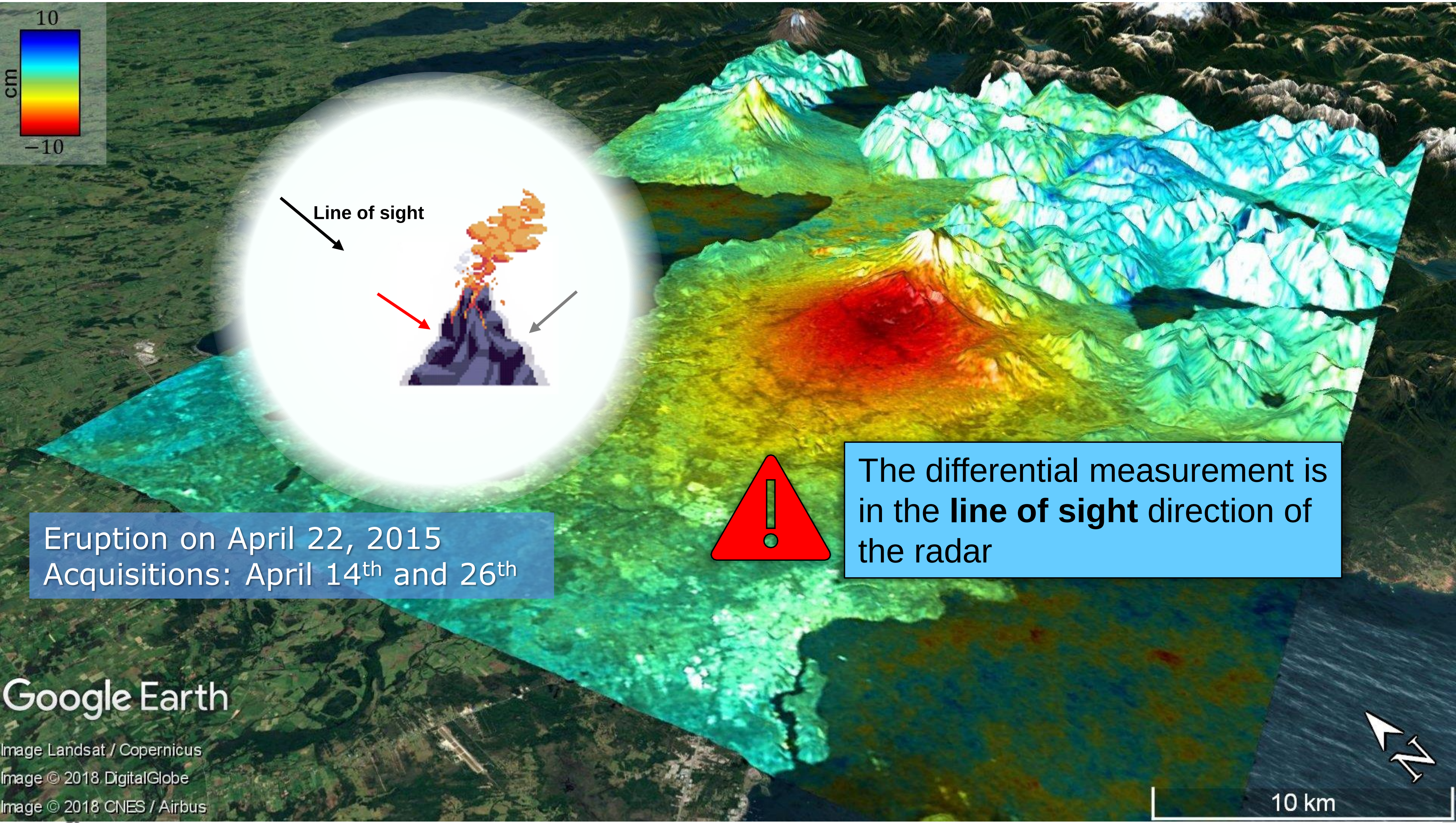
2015 Illapel Earthquake (Chile) Observed by Sentinel-1

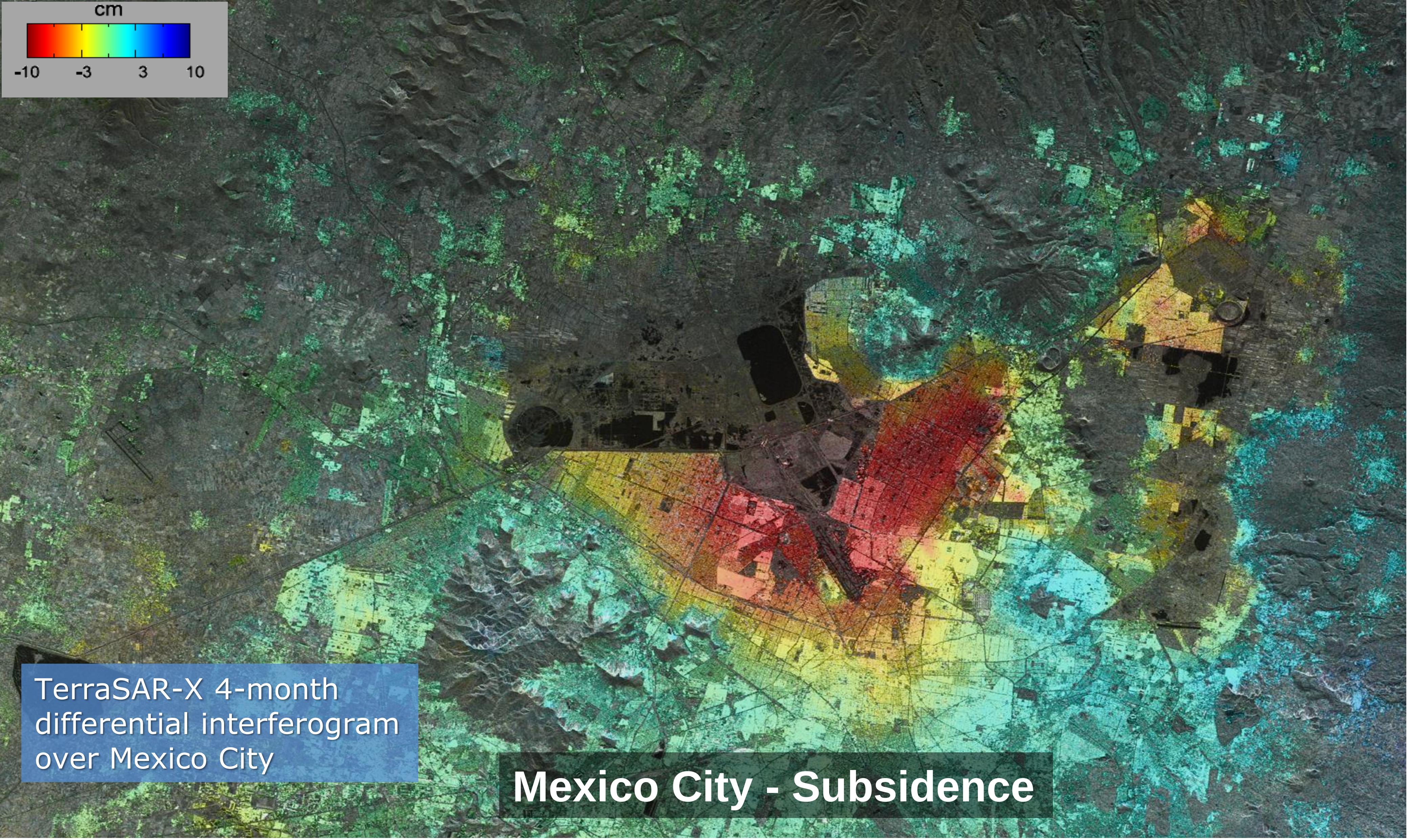


Calbuco Eruption Observed by Sentinel-1

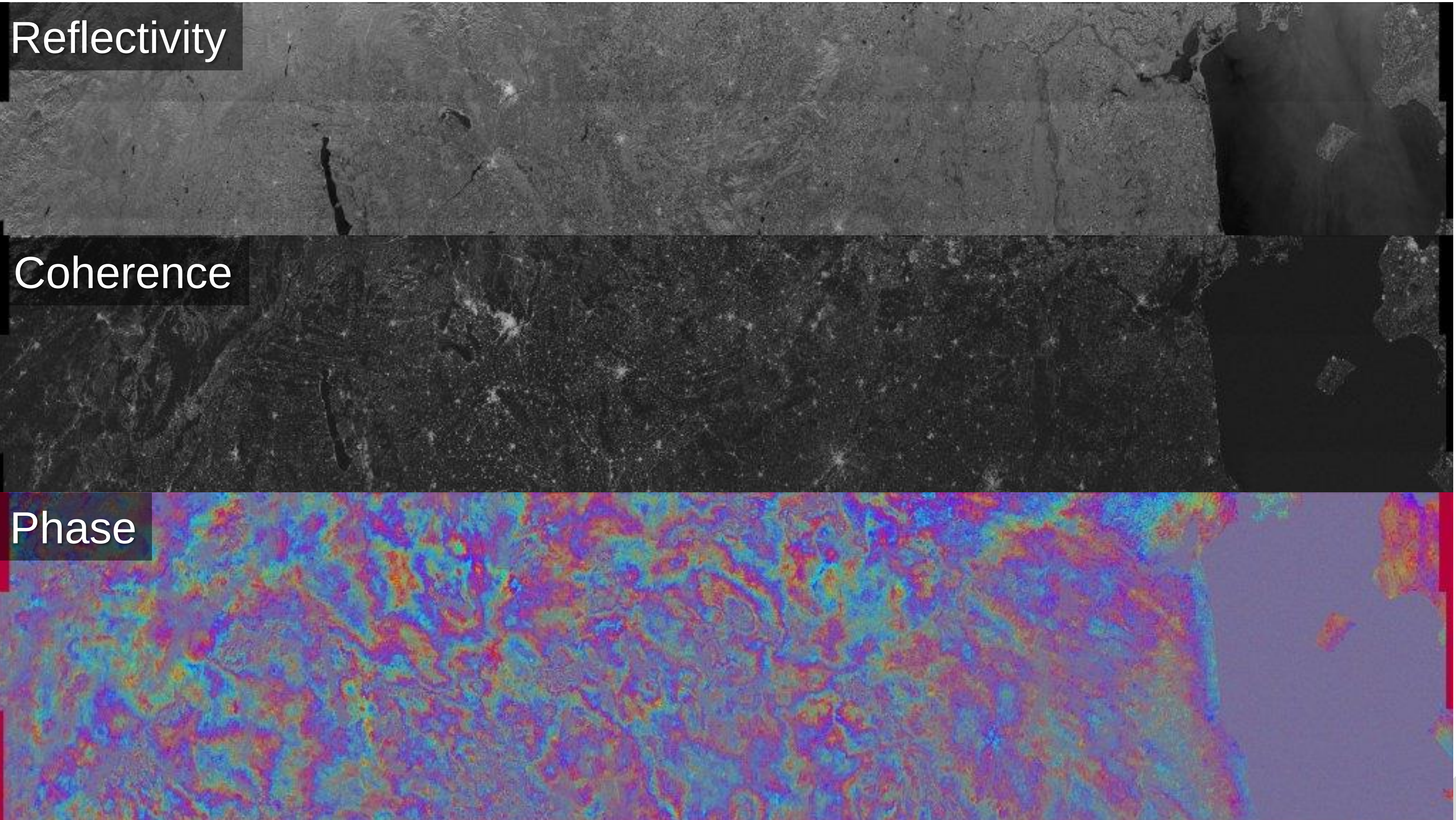


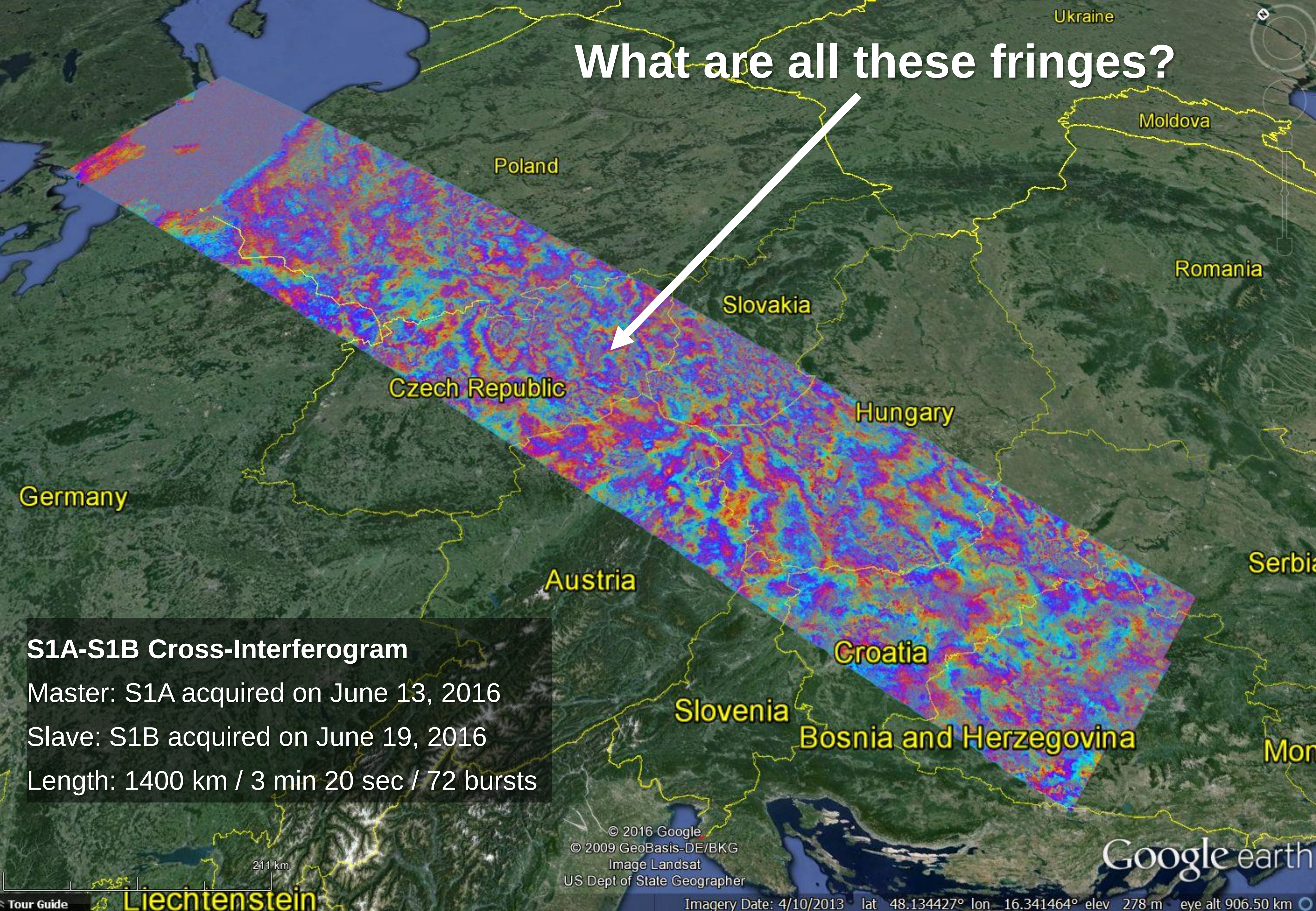
Calbuco Eruption Observed by Sentinel-1




cm
-10
-3
3
10
TerraSAR-X 4-month differential interferogram over Mexico City
Mexico City - Subsidence

Sentinel-1 Interferogram Over Europe (1400 km x 260 km)




What are all these fringes?
Ukraine
Moldova
Poland
Romania
Slovakia
Czech Republic
Hungary
Germany
Serbia
Austria
Croatia
Slovenia
Bosnia and Herzegovina
Liechtenstein
S1A-S1B Cross-Interferogram
Master: S1A acquired on June 13, 2016
Slave: S1B acquired on June 19, 2016
Length: 1400 km / 3 min 20 sec / 72 bursts
211 km
© 2016 Google
© 2009 GeoBasis-DE/BKG
Image Landsat
US Dept of State Geographer
Google earth
Tour Guide
Imagery Date: 4/10/2013 lat 48.134427° lon 16.341464° elev 278 m eye alt 906.50 km

Advanced Differential SAR Interferometry

- Main limitation of conventional differential SAR interferometry:
  - Reflectivity changes (coherence loss)
  - Atmosphere (mainly troposphere) biases the estimation
  - Residual unknown topography

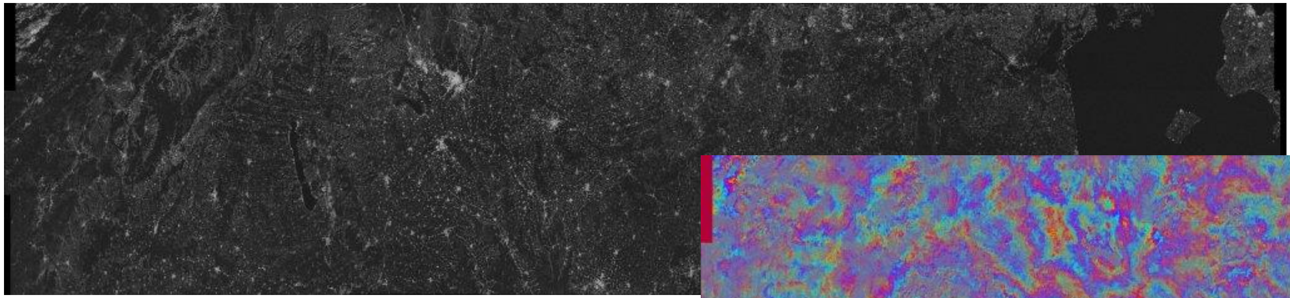
$$\varphi_{diff} = \varphi_{defo} + \varphi_{flat} + \varphi_{topo} + \varphi_{atmo} + \varphi_{noise}$$

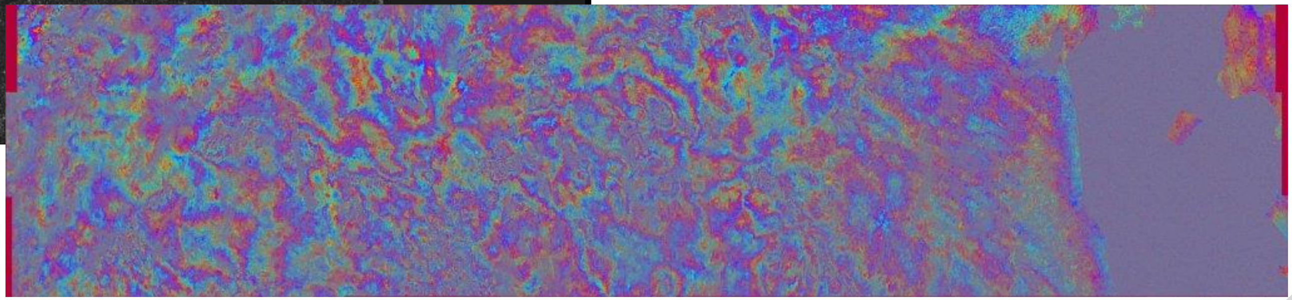
SNR dependent

Proportional to temporal baseline

Proportional to cross-track baseline

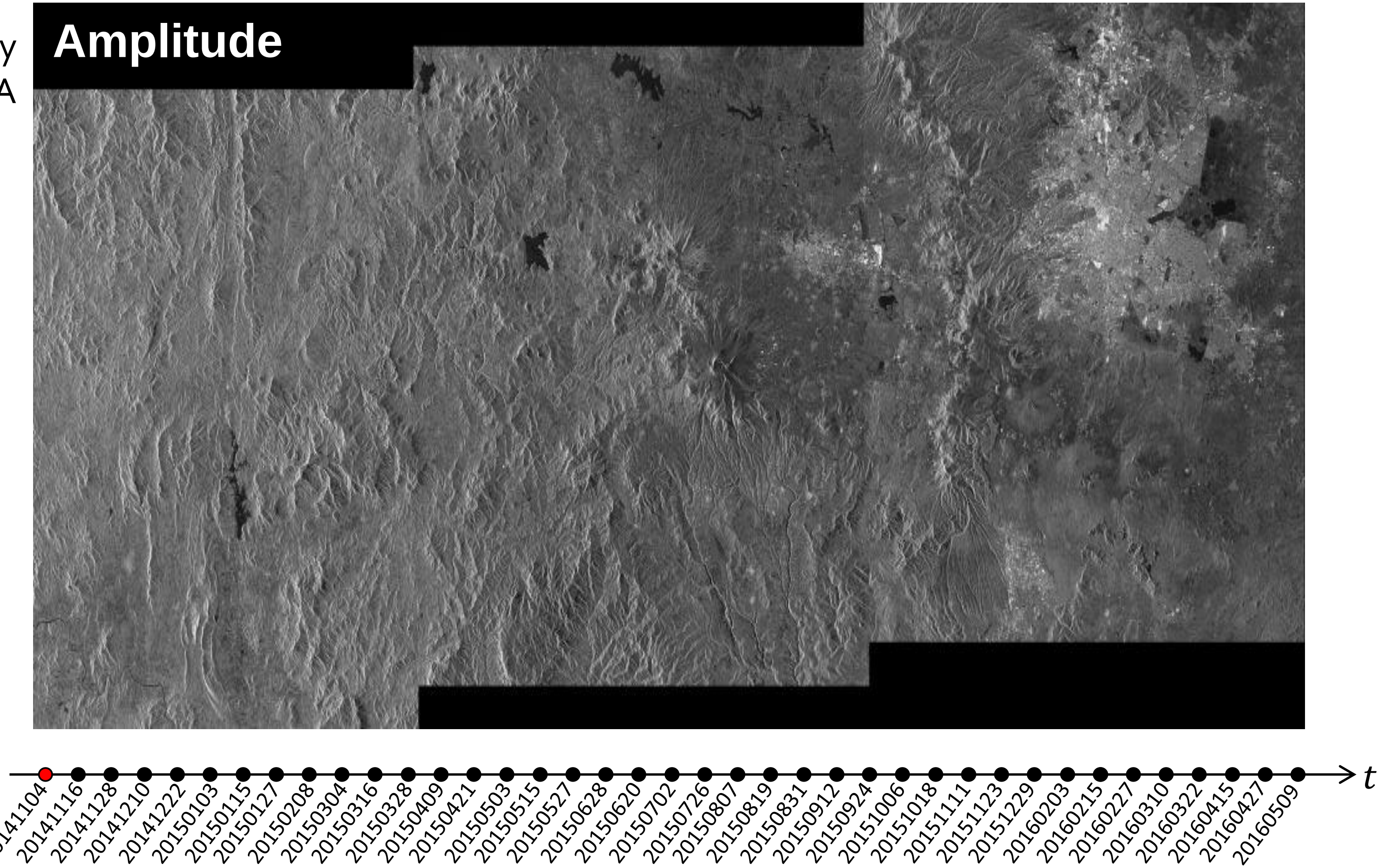
Phase due to atmospheric phase screen (variations)





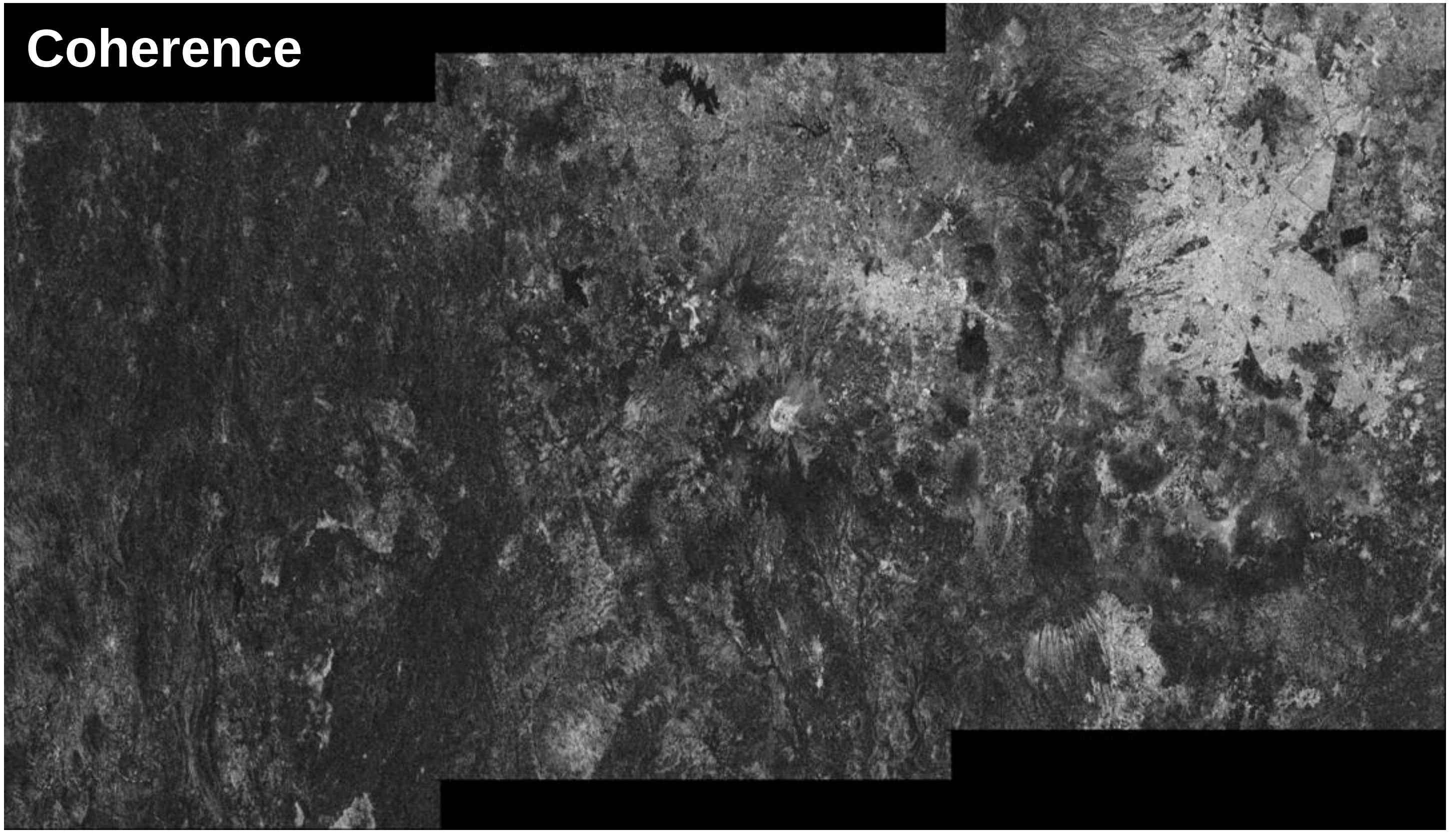
Temporal Evaluation of Interferometric Coherence

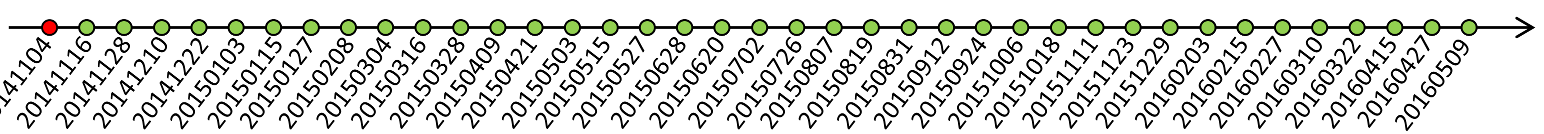
- Mexico City
- Sentinel-1A
- Ascending
- 18 months



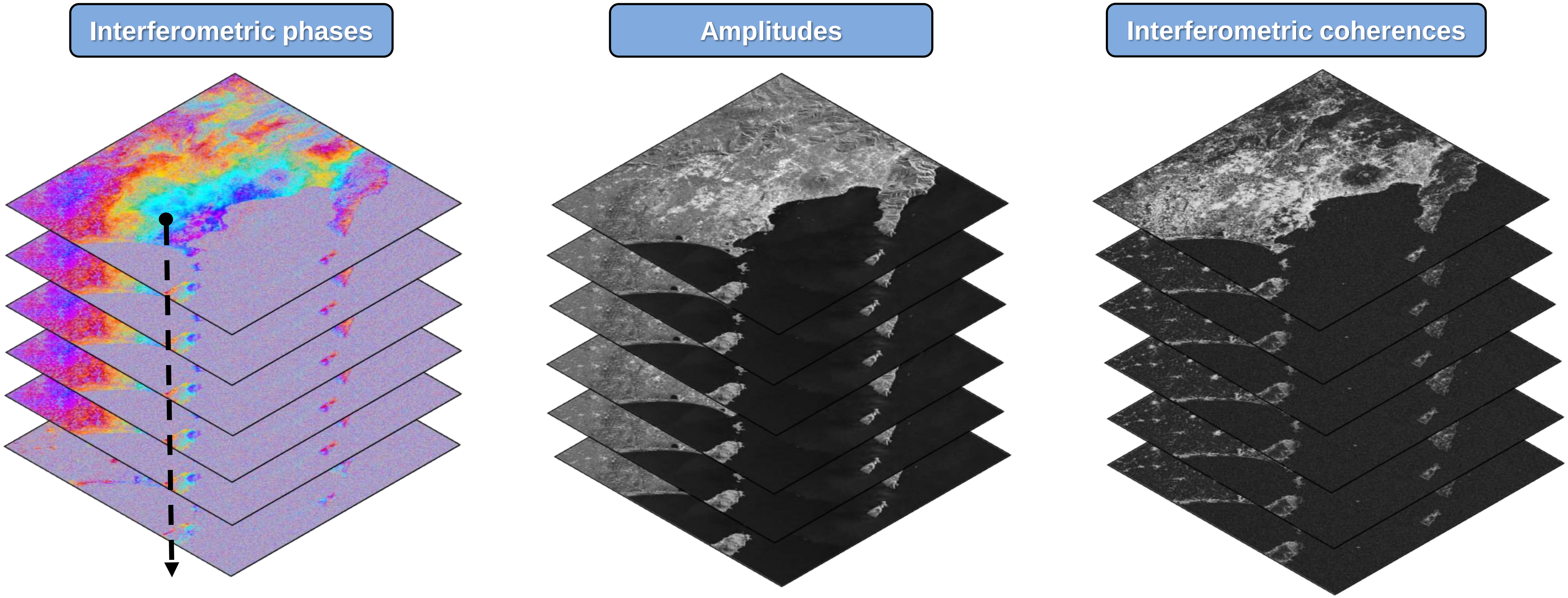
Temporal Evaluation of Interferometric Coherence

- Mexico City
- Sentinel-1A
- Ascending
- 18 months





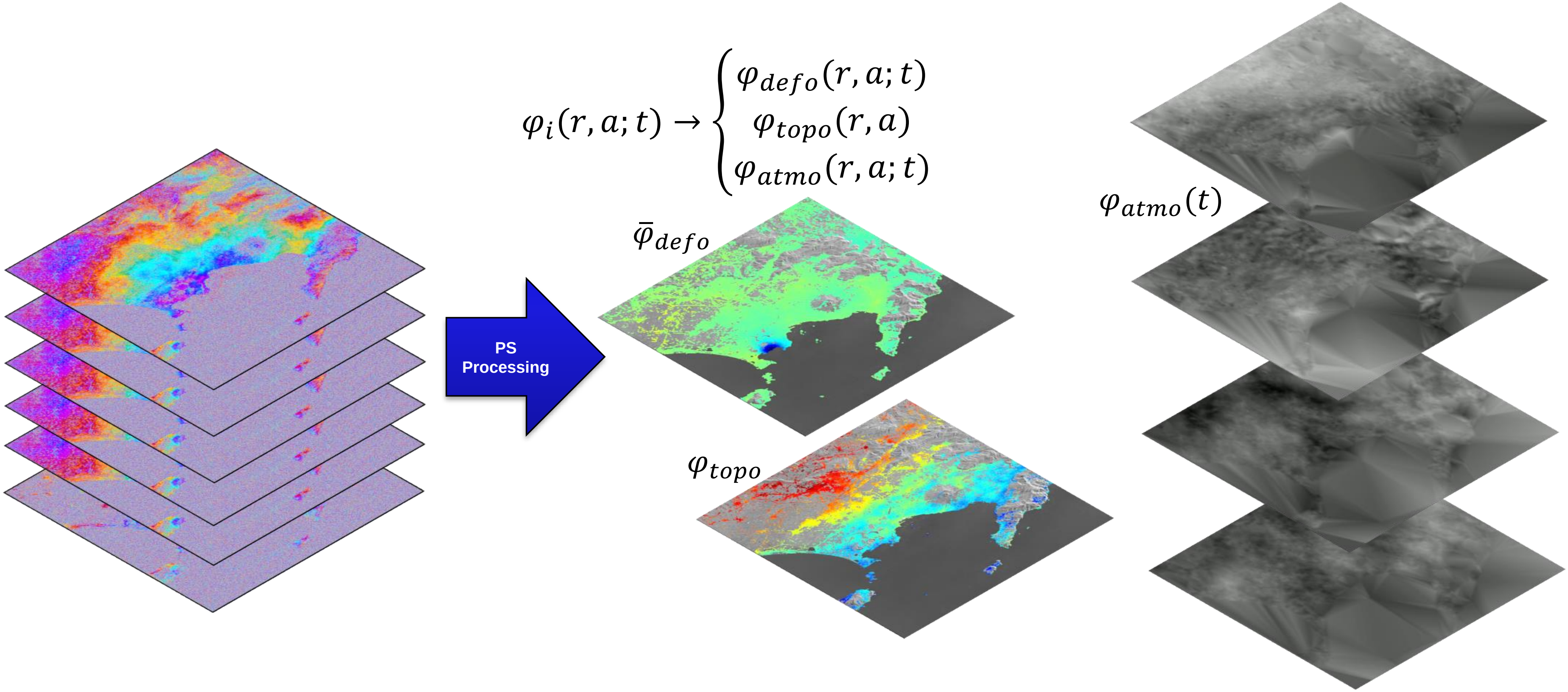
Exploitation of Time Series (Stacks)



- Exploitation of **image stacks** to retrieve accurate the **time series deformation**
- The measurement is obtained over the stable radar targets, which remain coherent in time: the so-called **permanent** or **persistent scatterers (PS)**
- Several techniques to select valid PS, e.g., based on **amplitude stability** or on **interferometric coherence**

Advanced DInSAR Processing

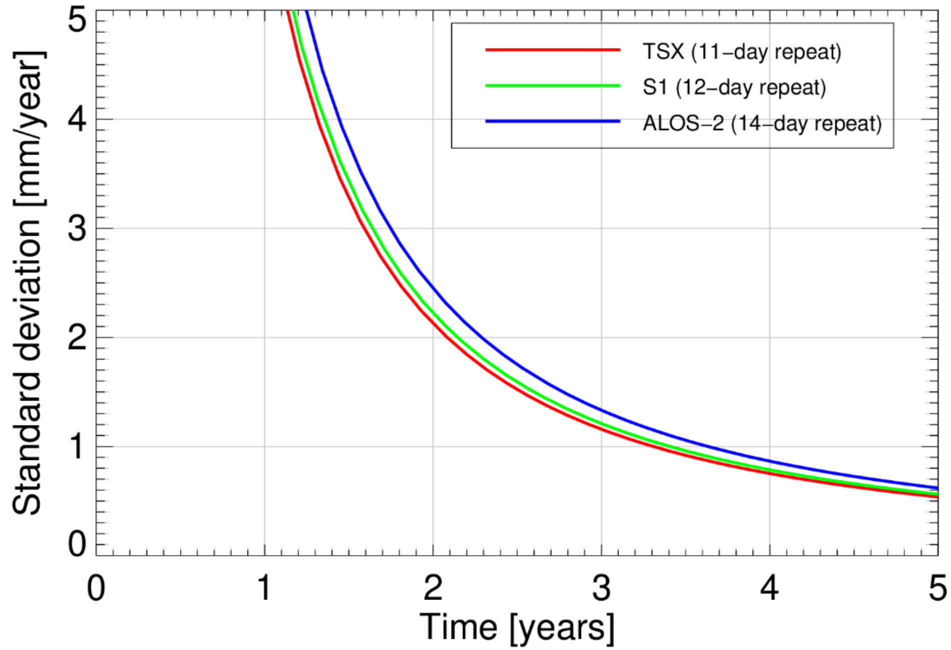
- Separation of the different contributions by means of advanced signal processing algorithms



- Many challenges: sparse grid of points, phase unwrapping, separation of atmosphere and motion, long time series, millions of points, etc.

Accuracy of DInSAR vs PSInSAR

- DInSAR (two images) performance is limited by atmospheric artifacts → $\sigma_{DInSAR} \approx 1$ cm
- PSInSAR performance depends on the **number of images** (span of time series) and **distance to calibration point** (its a *differential* measurement w.r.t. a reference point)



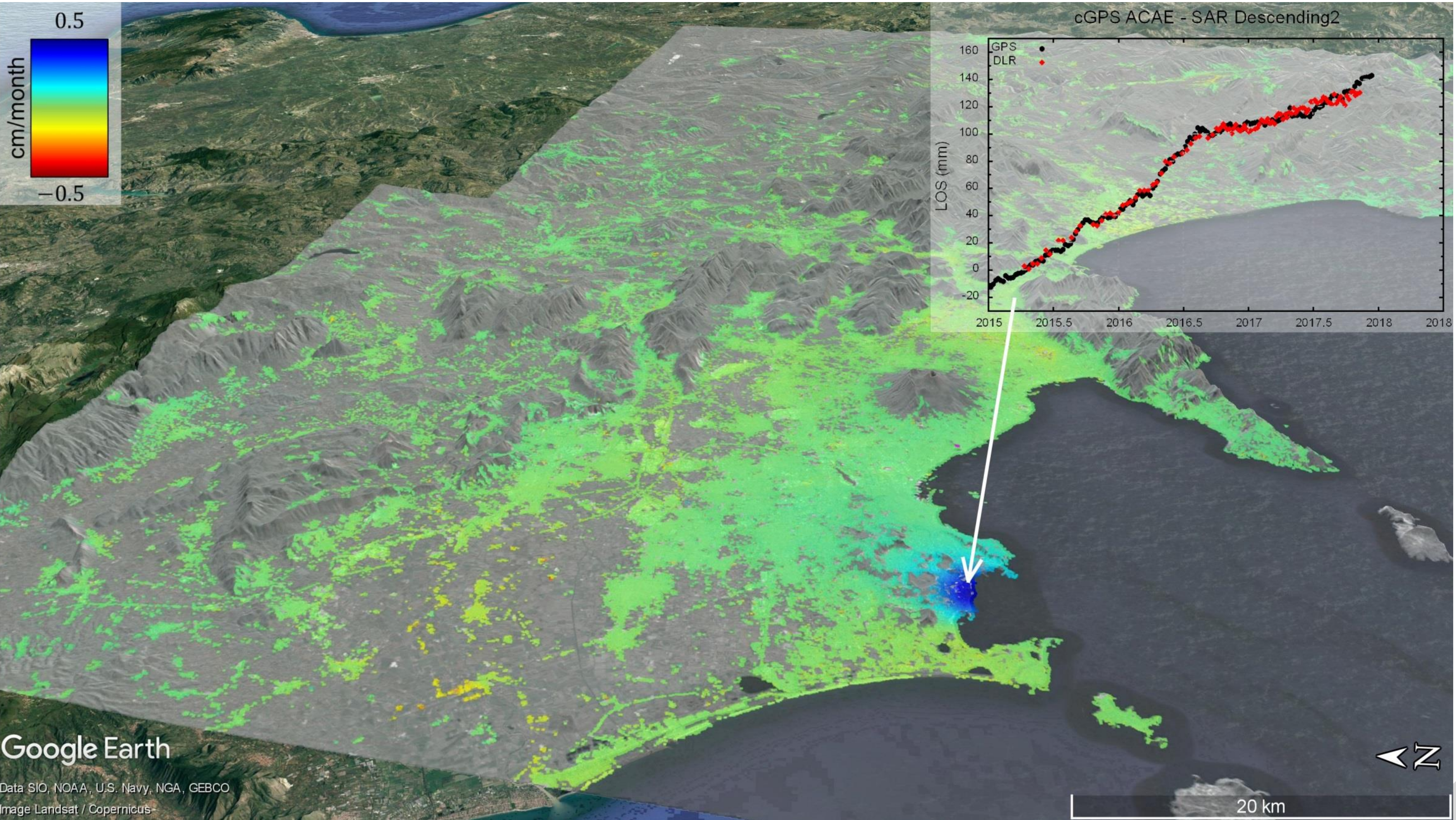
Performance depends on several assumptions, being the power of the troposphere the main one

- Smaller wavelength → more sensitive to motion

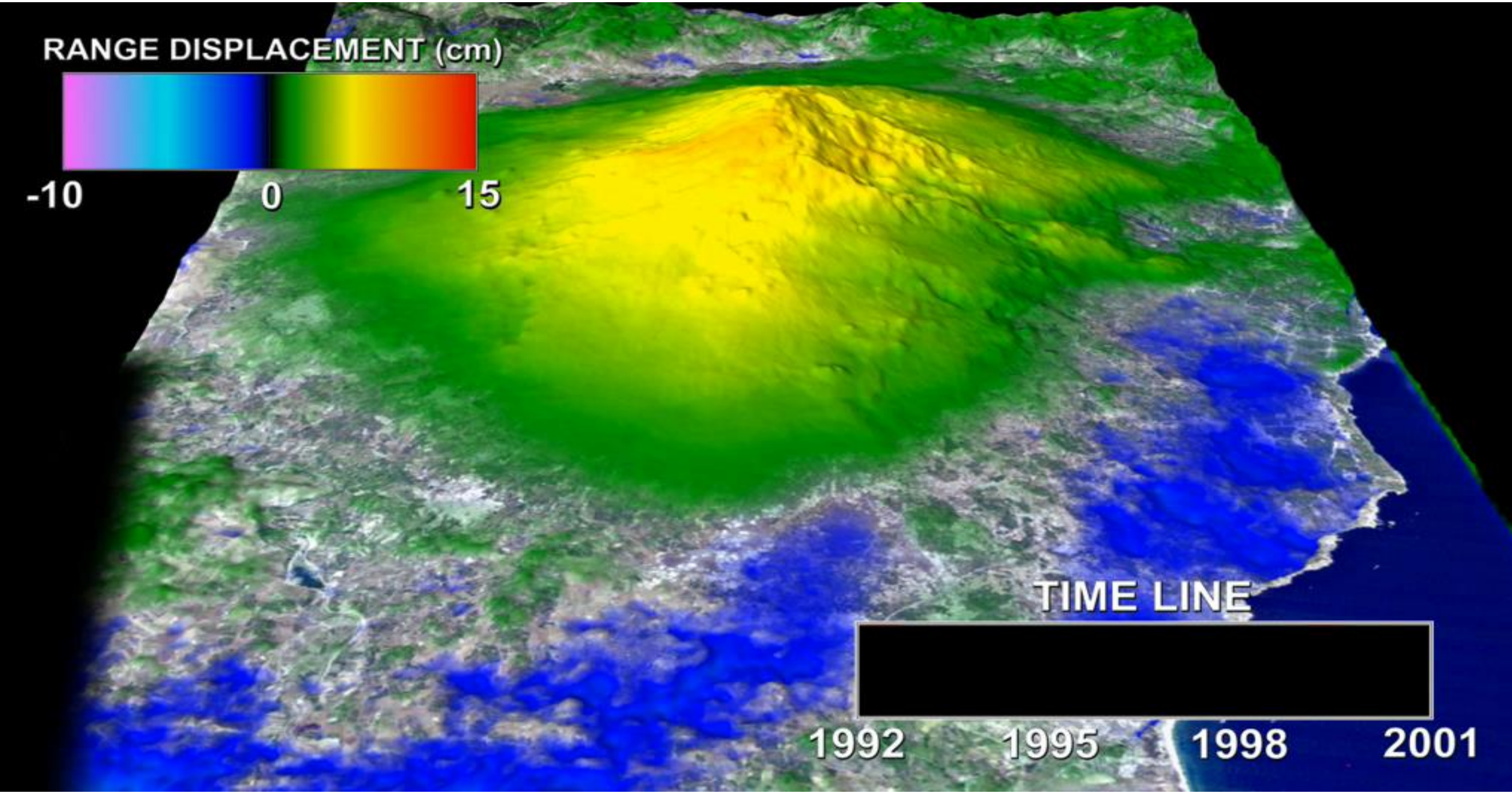
  …BUT also more sensitive to APS → similar performance at different wavelengths

  …BUT larger wavelengths have better long-term coherence → more PSs!

Time Series Deformation at the Campi Flegrei Caldera
Sentinel-1 A&B, 105 images, 2015-2017



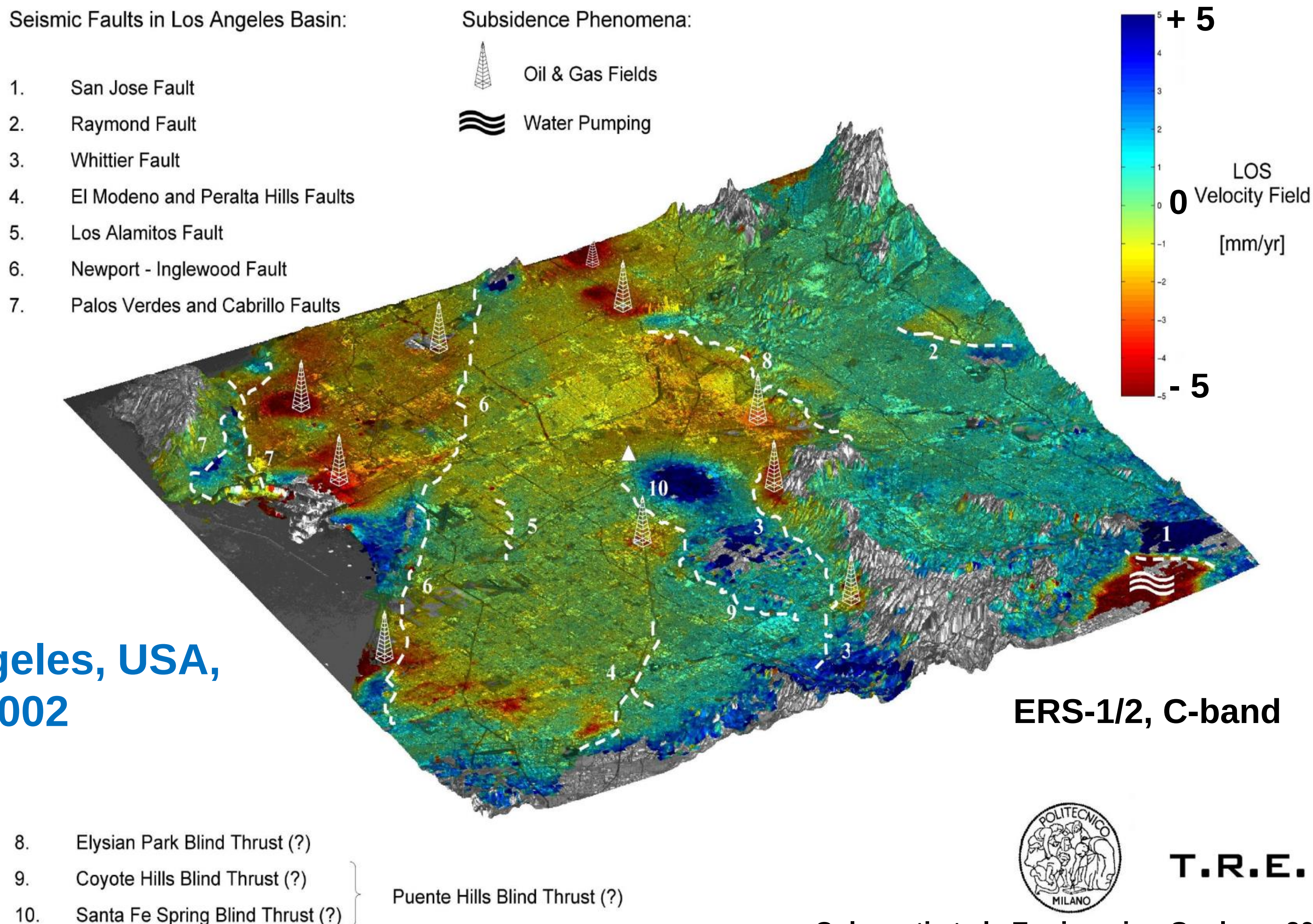
“Breathing” Etna



Etna Volcano, Italy (200 ERS-1/2 images)

(P. Lundgren, NASA JPL)

Subsidence due to oil extraction



Los Angeles, USA, 1992 - 2002

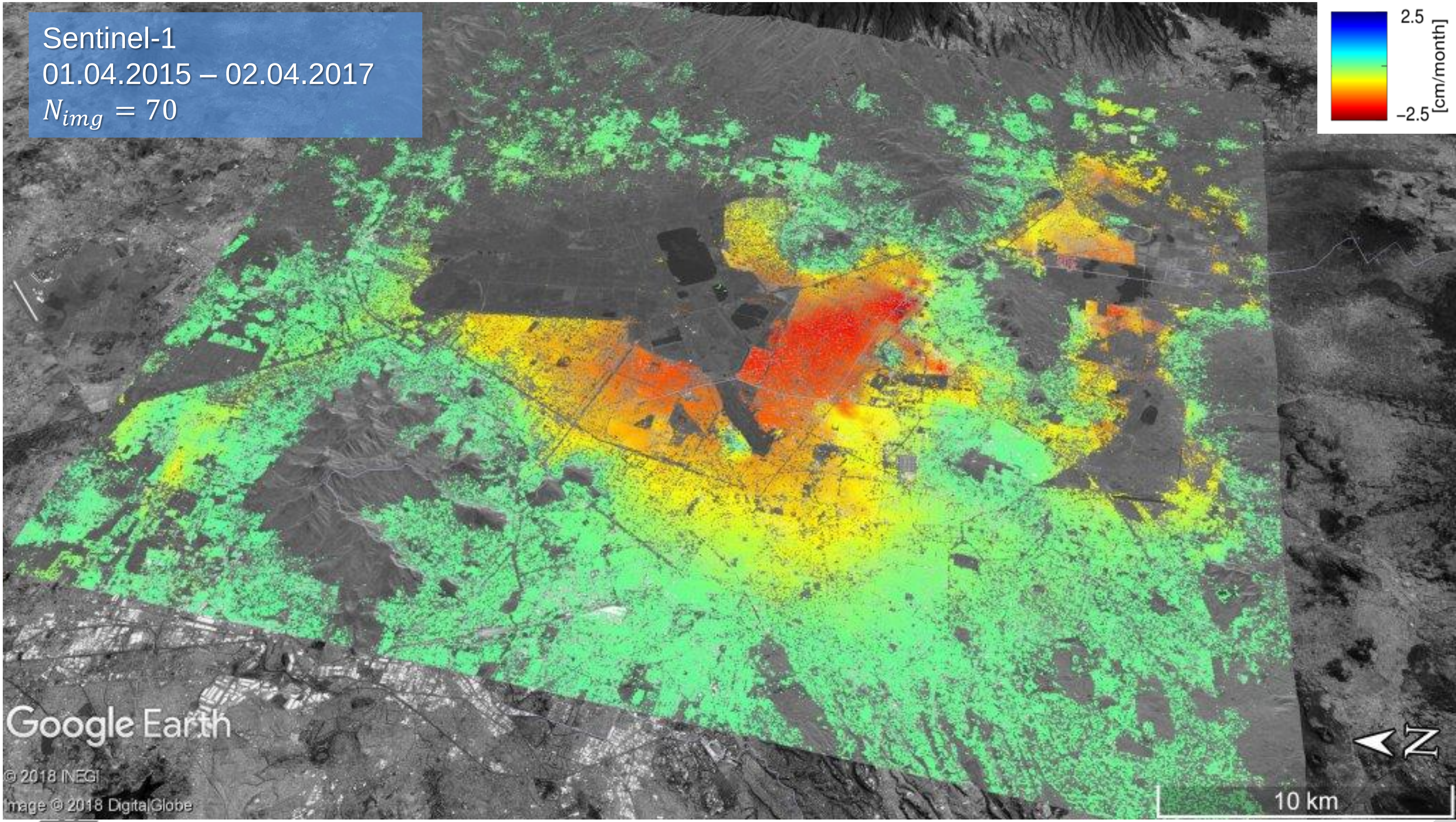
ERS-1/2, C-band

POLITECNICO MILANO

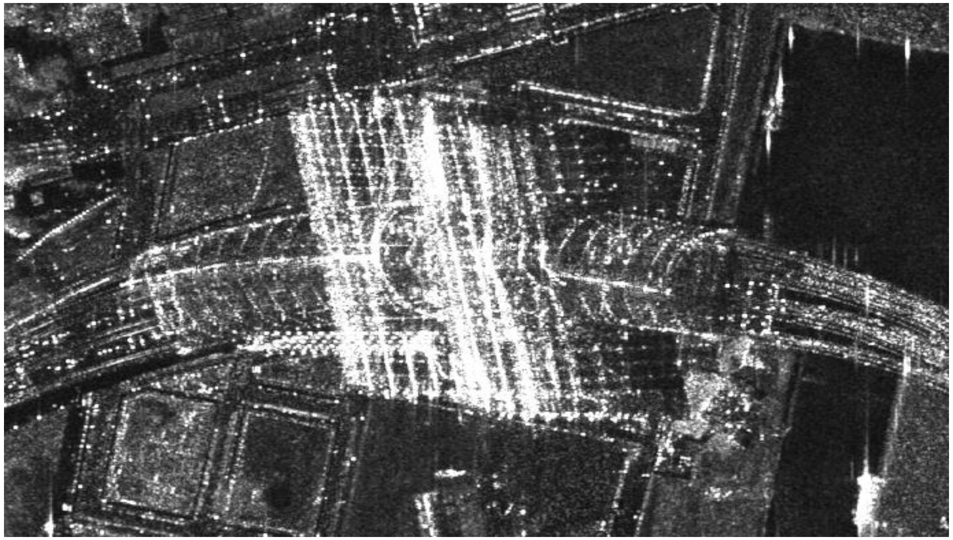
T.R.E.

Colesanti et al., Engineering Geology, 2003

Subsidence in Mexico City



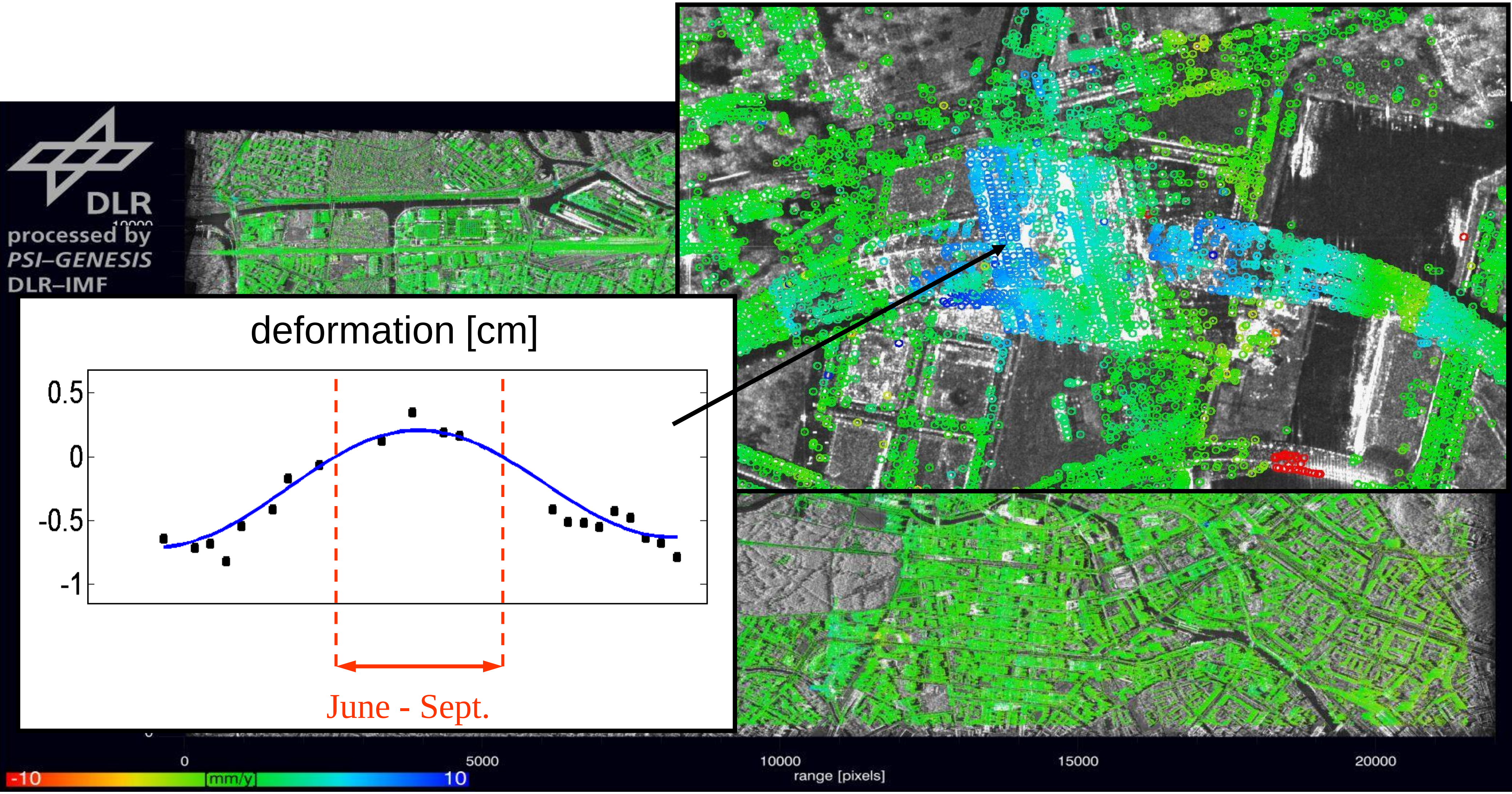
Deformation: Main Station Berlin



TerraSAR-X

Google Earth

Deformation: Main Station Berlin

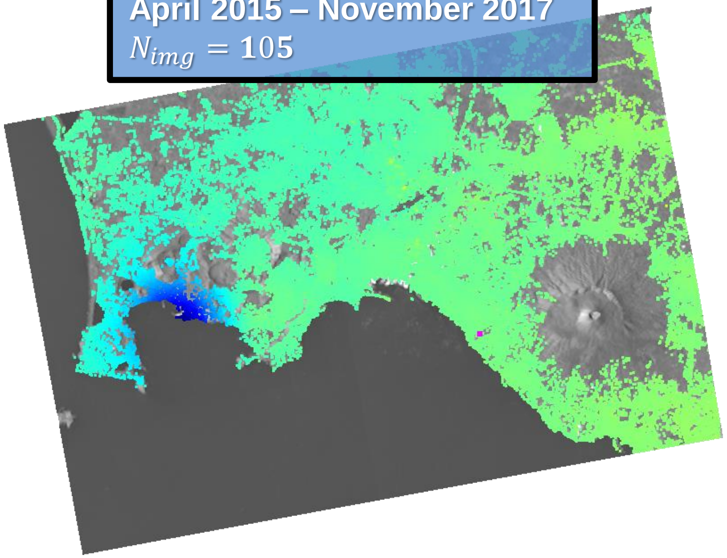


2D Deformation retrieval (ascending & descending)

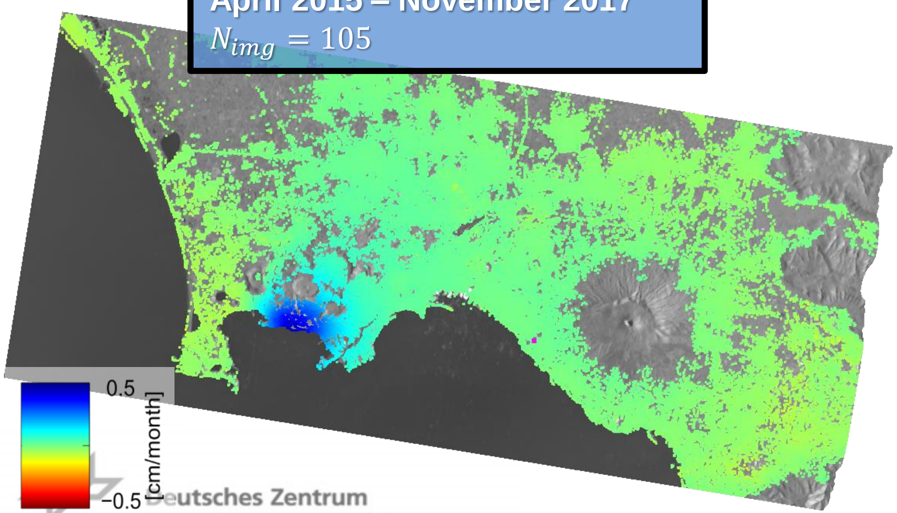
- Low Earth Orbit (LEO) SAR satellites have a quasi-polar orbit
- Almost insensitive to North-South motion
- East-West and Vertical motion can be retrieved by combining ascending and descending acquisitions
- Possibility to use external measurements, e.g., GNSS, to retrieve 3-D deformation (albeit with lower resolution in the N-S)

2D Deformation retrieval (ascending & descending)
Campi Flegrei Caldera, Italy, Sentinel-1 time series

Ascending
April 2015 – November 2017
$N_{img} = 105$

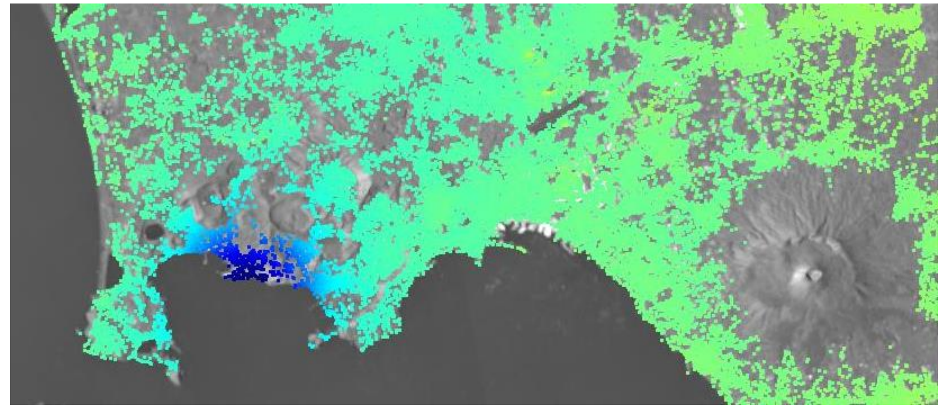


Descending
April 2015 – November 2017
$N_{img} = 105$

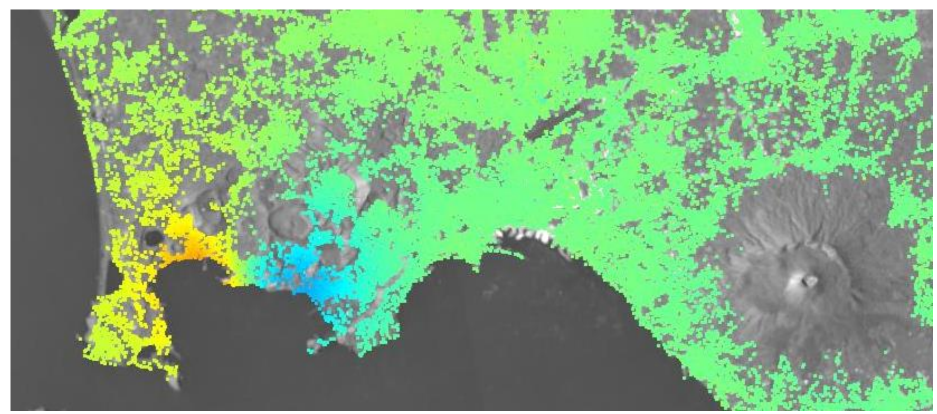


2D Inversion

Vertical

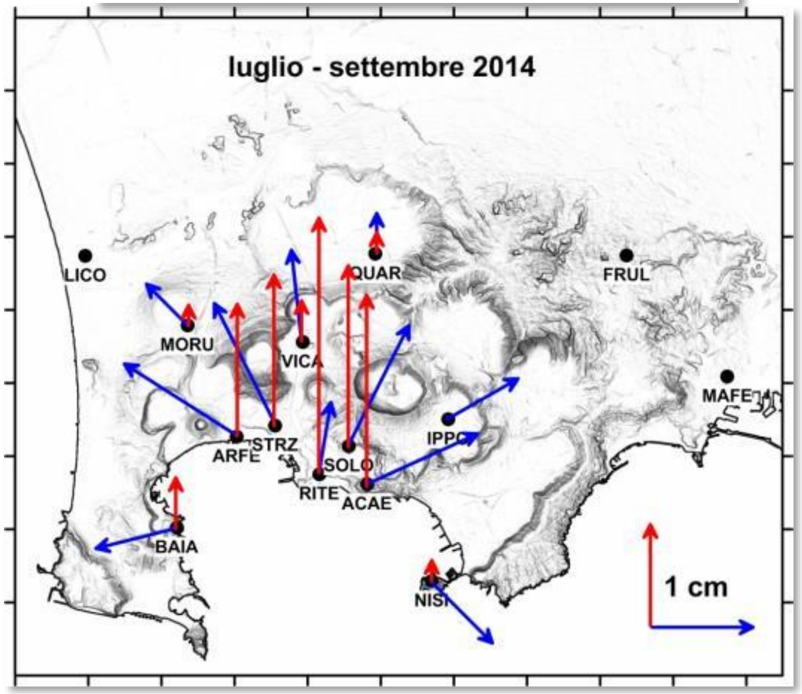


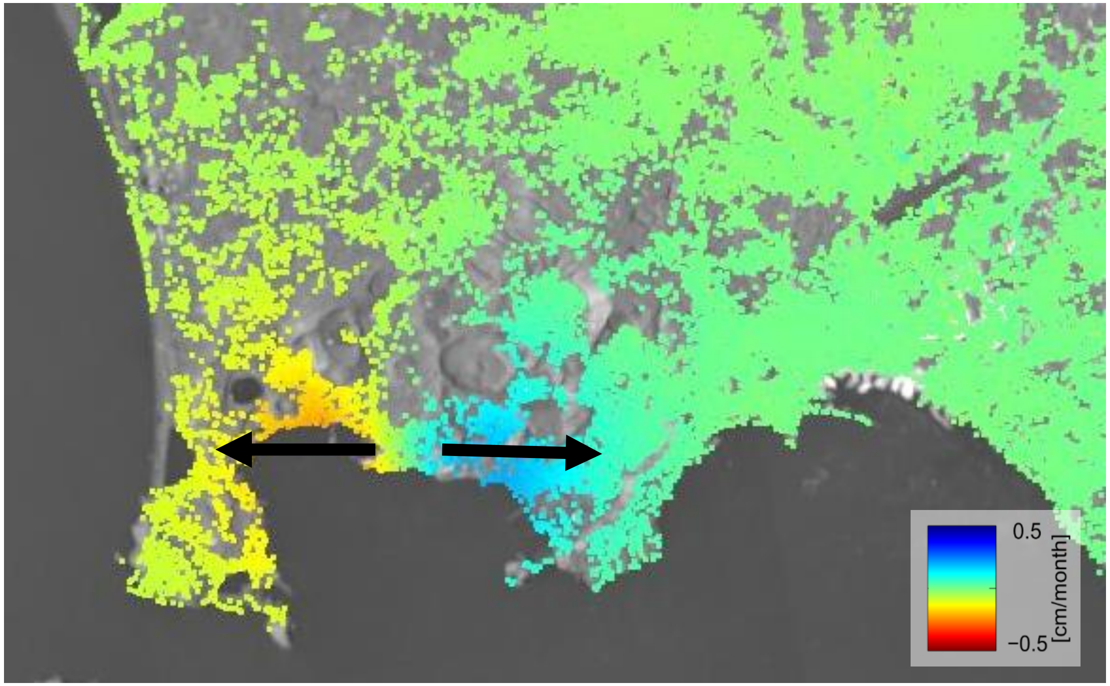
East-West



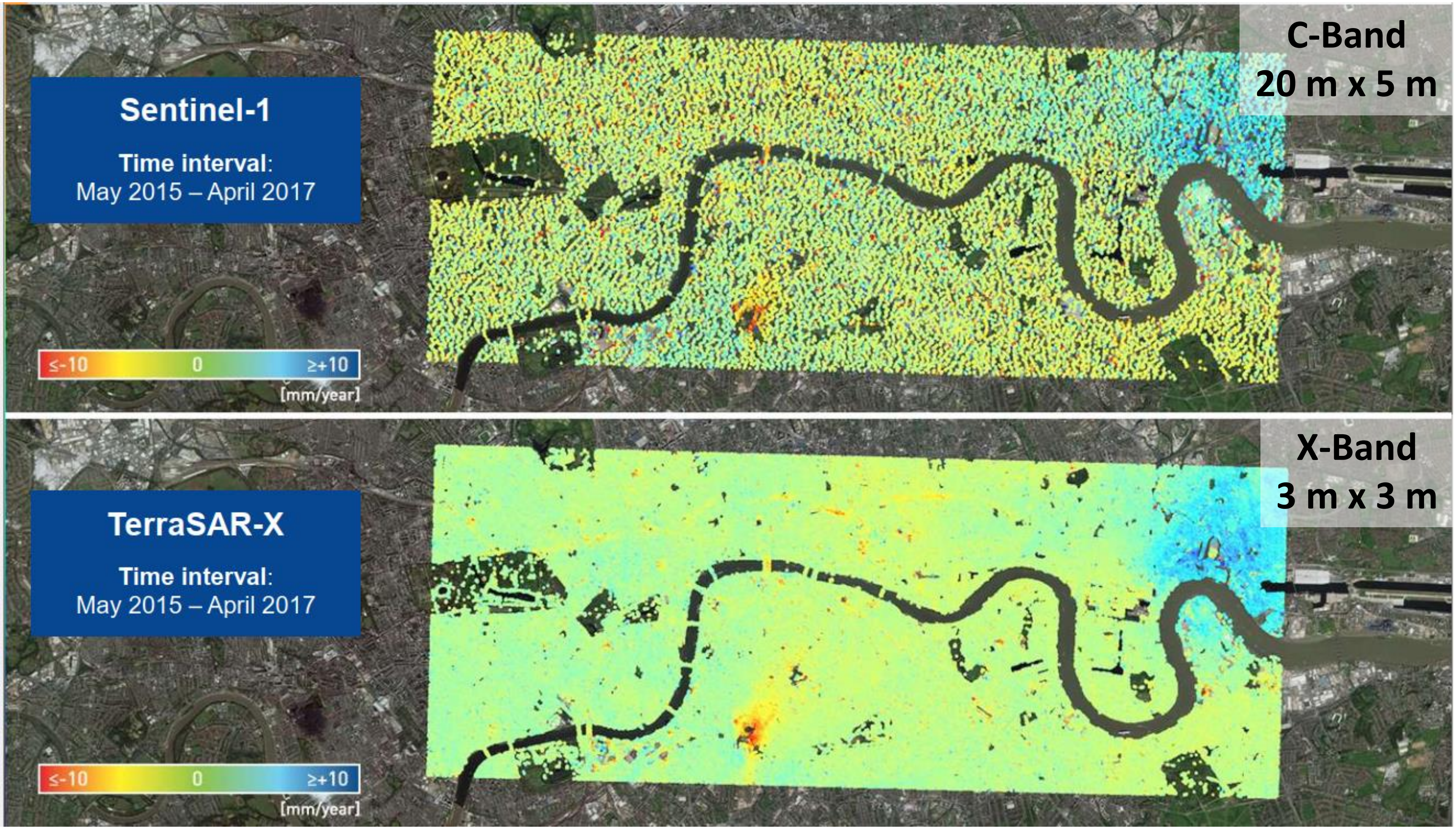
2D Deformation retrieval (ascending & descending) Campi Flegrei Caldera, Italy, Sentinel-1 time series

Non-negligible North component





The resolution factor

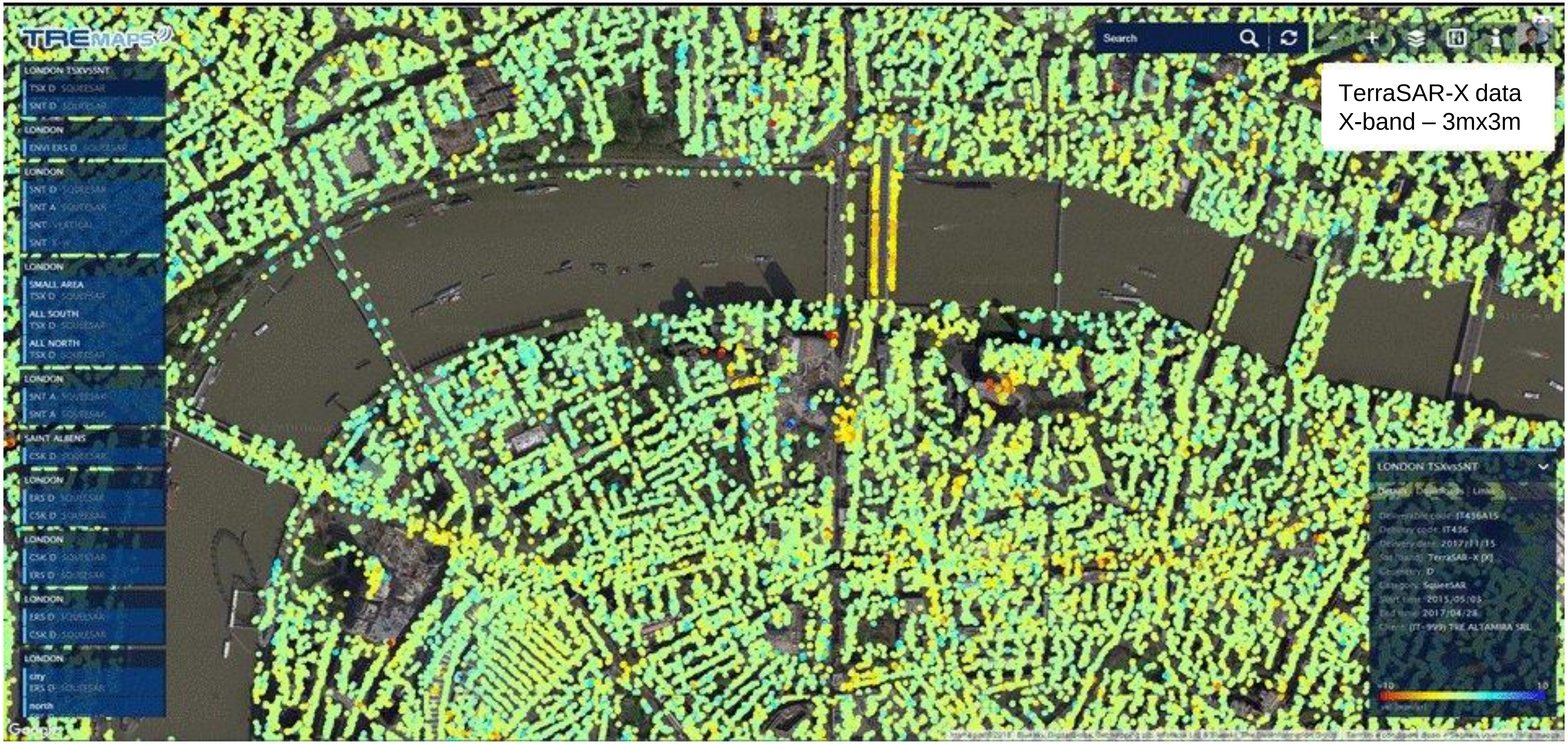


(A. Ferretti, TRE)

The resolution factor

(A. Ferretti, TRE)

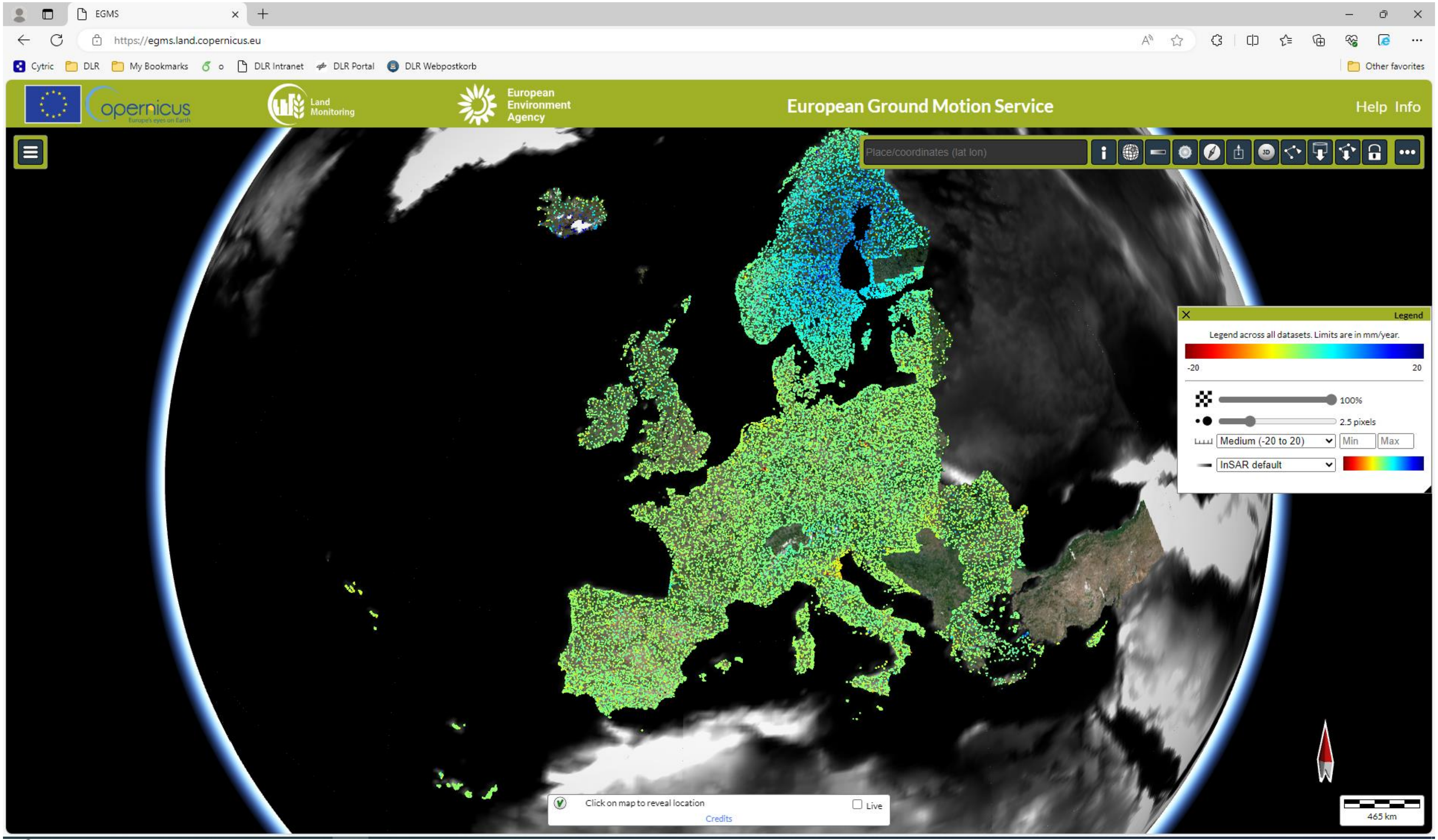
The resolution factor



(A. Ferretti, TRE)

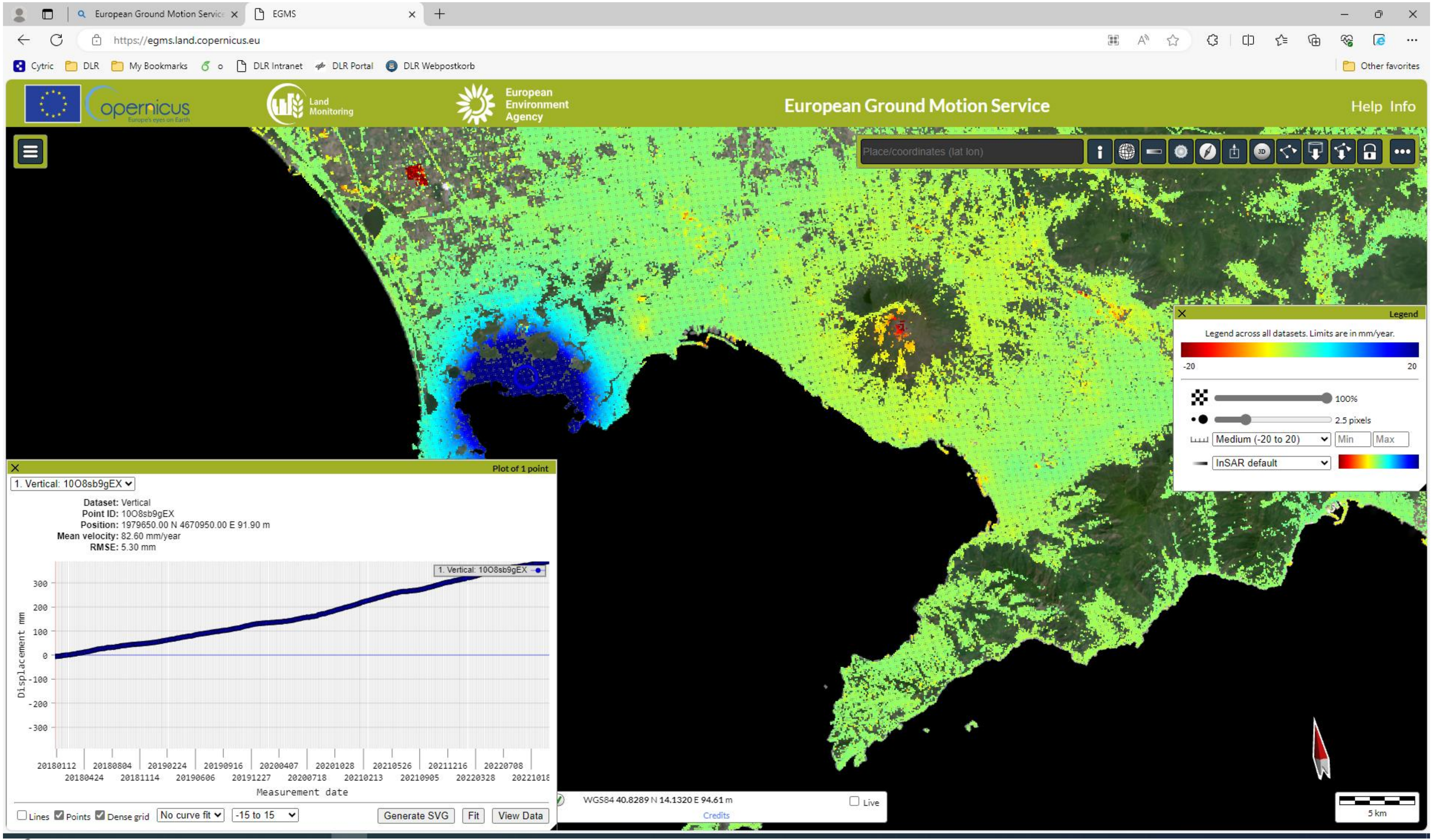
European Ground Motion Service (EGMS)

https://egms.land.copernicus.eu/



European Ground Motion Service (EGMS)

https://egms.land.copernicus.eu/



Software Tools for InSAR Processing

Public & Open Source

OpenSARlab, Alaska SAR facility (ASF, USA)	http://www.asf.alaska.edu/
Miami InSAR Time-series in Python (MintPy)	https://github.com/insarlab/MintPy
Doris (Delft University, The Netherlands)	http://doris.tudelft.nl/
Scientific Toolbox Exploitation Platform (ESA)	https://step.esa.int/main/
InSAR Scientific Computing Environment (JPL, USA)	https://isce-framework.github.io/isce3/
StaMPS/MTI (Uni Leeds, UK)	https://homepages.see.leeds.ac.uk/~earahoo/stamps/

SAR Interferometry Modes Summary

⇒ Two or more complex-valued SAR images are combined

⇒ Geometric information about the imaged objects (compared to using a single image) is derived by exploiting phase differences.

⇒ Images must differ in at least one aspect (= “baseline”)

Baseline type	Denomination	Application
$\Delta\theta$	**across-track**	**topography, Digital Elevation Models**
$\Delta t = \text{ms} \quad \text{to} \quad \text{s}$	**along-track**	**ocean currents, moving targets**
$\Delta t = \text{days}$	**differential**	**glacier/ ice fields / lava flows**
$\Delta t = \text{days} \ \text{to} \ \text{years}$	**differential**	**subsidence, seismic events volcanic activities, displacements**

The Pol-InSAR Course

Synthetic Aperture Radar (SAR)

Part 4: Differential InSAR

Prepared by DLR-HR's Pol-InSAR Team

German Aerospace Center (DLR), Microwaves & Radar Institute (HR), Pol-InSAR Research Group

Email: pau.prats@dlr.de, kristina.belinska@dlr.de, kostas.papathanassiou@dlr.de

DLR esa ESAMAAP EEBIOMASS